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Statistical process monitoring: basics and beyond

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This paper provides an overview and analysis of statistical process monitoring methods for fault detection, identification and reconstruction. Several fault detection indices in the literature are analyzed and unified. Fault reconstruction for both sensor and process faults is presented which extends the traditional missing value replacement method. Fault diagnosis methods that have appeared recently are reviewed. The reconstruction-based approach and the contribution-based approach are analyzed and compared with simulation and industrial examples. The complementary nature of the reconstruction- and contribution-based approaches is highlighted. An industrial example of polyester film process monitoring is given to demonstrate the power of the contribution- and reconstruction-based approaches in a hierarchical monitoring framework. Finally we demonstrate that the reconstruction-based framework provides a convenient way for fault analysis, including fault detectability, reconstructability and identifiability conditions, resolving many theoretical issues in process monitoring. Additional topics are summarized at the end of the paper for future investigation. **Copyright © 2003 John Wiley & Sons, Ltd.**

KEYWORDS: process monitoring; process chemometrics; fault detection; fault identification; fault reconstruction; sensor validation; contribution plots; fault analysis

1. INTRODUCTION

Process chemometrics applies multivariate analysis techniques to process data analysis and process improvements. An important area of tremendous success in process chemometrics is statistical process monitoring (SPM), which has become one of the most active research areas in process control over the last decade. Using methods from multivariate statistical analysis, SPM has found wide applications in different industrial processes, including chemicals, polymers, microelectronics manufacturing and pharmaceutical processes. The tasks involved in SPM typically include: (i) fault detection; (ii) fault identification and diagnosis; (iii) fault estimation, which assesses the fault magnitude; and (iv) fault reconstruction, which estimates the fault-free values to keep control and monitoring on-going even if some faults have occurred. Owing to the data-based nature of SPM, it is relatively easy to apply to real processes of rather large scale, in comparison with other methods based on systems theory or rigorous process models.

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While the process control community began to investigate the use of multivariate statistics for SPM in the late 1980s, [1–4], the use of multivariate statistics for abnormal situation detection has been studied intensively in the area of multivariate quality control (MQC) [5]. Typically, the Hotelling's T^2 statistic and the Q statistic, which is also known as the squared prediction error (SPE), are used for the detection of an out-of-control situation. These two statistics, typically calculated based on a model from principal component analysis (PCA) or partial least squares (PLS), give superior performance to the univariate quality control methods which monitor one variable at a time. The MQC literature, however, mainly focuses on the monitoring of quality variables and the *detection* of a quality problem, with few methods available for identifying root causes. The works of Kresta *et al.* [4] and Wise and co-workers [1, 6] are among the first to apply multivariate methods to process variables in addition to quality variables. Although these papers use virtually the same statistics as those used in MQC, such as the Q statistic and Hotelling's T^2 statistic, later process monitoring work extends the use of multivariate statistics for fault diagnosis and identification [7–17] and fault reconstruction [1, 6, 10, 12, 13].

Because statistical process monitoring focuses on process variables rather than just quality variables, the multivariate statistical models can actually extract the variable correlation due to mass balance, energy balance and other operational restrictions in an empirical way. We can summarize the

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following characteristics of process monitoring which distinguishes it from MQC: (i) fault diagnosis in process monitoring becomes feasible and more interesting owing to the inclusion of process variables in the analysis; (ii) fault reconstruction is possible based on multivariate statistical models to maintain control and optimization of process variables on-going even though some sensors have failed [6, 10, 15]; (iii) the stationarity assumption of multivariate statistical methods is challenged in SPM, since there are frequently normal process changes and process drifts which would be reflected in the process variables, while the quality variables are not supposed to change or drift—this leads to methods for multiscale and recursive monitoring to remove or adapt for process non-stationarity [18–22]; (iv) process dynamics becomes a concern, which causes autocorrelation in the variables [19, 23–25]; and (v) multiway analysis such as multiway PCA or multiway PLS is suitable for monitoring batch processes, which historically have less sophisticated control strategies than their continuous counterparts [26].

The tasks in process monitoring can be compared in parallel to those in the following areas: (i) gross error detection and identification (GDI) based on first-principles models [27–30]; (ii) fault detection and isolation (FDI) based on FDI observers (a special form of observers), parity relations and Kalman filters [31–36]; and (iii) multivariate statistics-based outlier detection and missing value replacement [5, 37–39]. The multivariate process monitoring methods based on PCA and PLS models offer a practical approach for fault detection and diagnosis. While fault detection is accomplished by directly applying statistics used in MQC, fault diagnosis is made possible by the use of contribution plots [8, 9, 40, 41] and a fault identification index based on fault reconstruction [10, 13]. Recent work by Gertler *et al.* [16] describes an isolation-enhanced PCA approach which uses a bank of PCA models for fault identification. A structured residuals approach with maximized sensitivity for fault diagnosis in processes is proposed by Qin and Li [15]. A related data-based method is the use of auto-associative neural networks as non-linear PCA for sensor validation [42]. In these methods, quasi-steady state models are used to detect sensor gross errors.

While the area of statistical process monitoring has progressed rapidly, with many successful industrial applications reported, only a few efforts have been made to provide overviews of the area. MacGregor [43] and MacGregor and Kourti [44] provide early overviews of the methods available in SPM. Wise and Gallagher [45] provide an overview of many aspects of statistical process monitoring based on PCA, PLS and their variations. A recent text by Chiang *et al.* [46] on process monitoring provides an introduction to SPM methods and their applications.

The objective of this paper is to provide an overview and analysis of recently developed process monitoring methods for fault detection, reconstruction and identification. The focus of the review necessarily reflects the author's experience in this area. Methods that are well reviewed in earlier papers [44, 45] will be mentioned here but will not be repeated at length. The paper is organized as follows. Section 2 discusses how processes are modeled using statistical methods and how faults are represented using subspaces or directions. Section 3 summarizes many existing fault detection indices, including global and subspace-based indices. A unified representation of these indices is presented as well. Section 4 discusses methods for fault reconstruction and their relation to missing value replacement approaches. Fault diagnosis methods are summarized in Section 5, with special attention to reconstruction-based methods and contribution plot approaches. Section 6 provides an industrial application in which the contribution- and reconstruction-based methods are used in a hierarchical monitoring framework. The analysis of fault detectability, reconstructability and identifiability is given in Section 7. Section 8 gives conclusions and further discussion.

2. PROCESS AND FAULT MODELING

Statistical process monitoring relies on the use of normal process data to build process models. These models include from PCA, PLS and their variants. PCA models are predominantly used to extract variable correlation from data [4, 6]. Wise and co-workers [6, 47] suggest the use of PLS models for process monitoring in a similar manner to PCA models, but they point out different characteristics of the two types of models. In this section we discuss the main points of SPM using PCA models, because PLS has been used in a similar manner.

2.1. Principal component analysis

Let $\mathbf{x} \in \mathbb{R}^m$ denote a sample vector of m sensors. Assuming that there are N samples for each sensor, a data matrix $\mathbf{X} \in \mathbb{R}^{N\times m}$ is composed with each row representing a sample. The matrix $\mathbf{X}$ is scaled to zero mean for covariance-based PCA and, in addition, to unit variance for correlation-based PCA. The matrix $\mathbf{X}$ can be decomposed into a score matrix $\mathbf{T}$ and a loading matrix $\mathbf{P}$ by either the NIPALS [48] or the singular value decomposition (SVD) algorithm:

$$\begin{aligned}\mathbf{X} &= \mathbf{T}\mathbf{P}^{\mathrm{T}} + \tilde{\mathbf{X}} = \mathbf{T}\mathbf{P}^{\mathrm{T}} + \tilde{\mathbf{T}}\tilde{\mathbf{P}}^{\mathrm{T}} \\ &= [\mathbf{T}\ \ \tilde{\mathbf{T}}]\,[\mathbf{P}\ \ \tilde{\mathbf{P}}]^{\mathrm{T}} \equiv \bar{\mathbf{T}}\bar{\mathbf{P}}^{\mathrm{T}}\end{aligned} \tag{1}$$

where $\tilde{\mathbf{X}} = \tilde{\mathbf{T}}\tilde{\mathbf{P}}^{\mathrm{T}}$ is the residual matrix, $\bar{\mathbf{T}} = [\mathbf{T}\ \ \tilde{\mathbf{T}}]$ and $\bar{\mathbf{P}} = [\mathbf{P}\ \ \tilde{\mathbf{P}}]$. Since the columns of $\bar{\mathbf{T}}$ are orthogonal, the covariance matrix is

$$\mathbf{S} \approx \frac{1}{N-1}\mathbf{X}^{\mathrm{T}}\mathbf{X} = \bar{\mathbf{P}}\bar{\mathbf{\Lambda}}\bar{\mathbf{P}}^{\mathrm{T}} \tag{2}$$

where

$$\bar{\mathbf{\Lambda}} = \frac{1}{N-1}\bar{\mathbf{T}}^{\mathrm{T}}\bar{\mathbf{T}} = \mathrm{diag}\{\lambda_1, \lambda_2, \ldots, \lambda_m\} \tag{3}$$

and

$$\lambda_i = \frac{1}{N-1}\mathbf{t}_i^{\mathrm{T}}\mathbf{t}_i \approx \mathrm{var}\{\mathbf{t}_i\} \tag{4}$$

when N is very large. The score vector $\mathbf{t}_i$ is the ith column of $\bar{\mathbf{T}}$, and λ_i are the eigenvalues of the covariance matrix in descending order. For variance-scaled $\mathbf{X}$, Equation (2) gives the correlation matrix $\mathbf{R}$. The principal component subspace (PCS) is $\mathcal{S}_{\mathrm{p}} = \mathrm{span}\{\mathbf{P}\}$ and the residual subspace (RS) is $\mathcal{S}_{\mathrm{r}} = \mathrm{span}\{\tilde{\mathbf{P}}\}$.

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A sample vector $\mathbf{x}$ can be projected on the PCS and RS respectively:

$$\hat{\mathbf{x}} = \mathbf{P}\mathbf{P}^{\mathrm{T}}\mathbf{x} \equiv \mathbf{C}\mathbf{x} \in \mathcal{S}_{\mathrm{p}} \tag{5}$$

$$\tilde{\mathbf{x}} = \tilde{\mathbf{P}}\tilde{\mathbf{P}}^{\mathrm{T}}\mathbf{x} = (\mathbf{I} - \mathbf{P}\mathbf{P}^{\mathrm{T}})\,\mathbf{x} = (\mathbf{I} - \mathbf{C})\,\mathbf{x} \in \mathcal{S}_{\mathrm{r}} \tag{6}$$

Since $\mathcal{S}_{\mathrm{p}}$ and $\mathcal{S}_{\mathrm{r}}$ are orthogonal,

$$\hat{\mathbf{x}}^{\mathrm{T}}\tilde{\mathbf{x}} = 0 \tag{7}$$

and

$$\mathbf{x} = \hat{\mathbf{x}} + \tilde{\mathbf{x}} \tag{8}$$

2.2. Fault direction matrix

The sample vector for normal operating conditions is denoted by $\mathbf{x}^*$, which is unknown when a fault has occurred. In the presence of a process fault $\mathcal{F}_i$ the sample vector $\mathbf{x}$ is represented by the expression

$$\mathbf{x} = \mathbf{x}^* + \Xi_i\mathbf{f} \tag{9}$$

where Ξ_i is orthonormal and $||\mathbf{f}||$ represents the magnitude of the fault. Note that $\mathbf{f}$ may change over time depending on how the actual fault develops over time. The actual fault belongs to the set of possible faults, denoted by $\{\mathcal{F}_i\}$. Some members of this set may be combinations of faults. For unidimensional faults, Ξ_i is a column vector, while for multidimensional faults, Ξ_i is a matrix. It is straightforward to derive Ξ_i for sensor faults. For example,

$$\Xi_i^{\mathrm{T}} = [0 \;\; 0 \;\; 1 \;\; 0 \;\; \cdots] \tag{10}$$

represents a single sensor fault in sensor 3, while

$$\Xi_i^{\mathrm{T}} = \begin{bmatrix} 1 & 0 & 0 & 0 & \cdots \\ 0 & 0 & 1 & 0 & \cdots \end{bmatrix} \tag{11}$$

represents a simultaneous sensor fault in sensors 1 and 3.

For process faults, Ξ_i represents the direction or subspace in which the process deviates from the normal situation. Process faults are usually multidimensional, but they can be unidimensional and at the same time impact multiple sensors. Examples of such can be found in e.g. Reference [13].

Yoon and MacGregor [49] classify faults into simple faults and complex faults. It should be noted that Ξ_i can represent both simple and complex faults. If faulty data are available for a complex process fault of interest, the fault direction matrix Ξ_i can be extracted from faulty data. Valle-Cervantes *et al*. [50] and Yue and Qin [51] give examples of extracting process fault directions for both continuous and batch processes.

2.3. How does PCA model a process?

In statistical process monitoring, PCA is used to model the correlation among process variables. Therefore it is desirable to understand what PCA is actually modeling about the process and how good the model is. Industrial processes usually have many unmeasured but normal process disturbances. We can represent the process at quasi-steady state as

$$\begin{aligned} \mathbf{B}\bar{\mathbf{x}}(k) + \mathbf{D}\mathbf{d}(k) &= \mathbf{0} \\ \mathbf{x}(k) &= \bar{\mathbf{x}}(k) + \mathbf{n}(k) \end{aligned} \tag{12}$$

where $\bar{\mathbf{x}}(k)$ represents the noise-free values, $\mathbf{d}(k)$ is the unmeasured disturbances and $\mathbf{n}(k)$ is the measured noise. The first part of Equation (12) is a representation of the process conservation laws at quasi-steady state. The second part of Equation (12) is the measurement equation. The time index k can be suppressed for convenience.

Denoting $\mathbf{D}^{\perp}$ as the orthogonal complement of $\mathbf{D}$ such that $\mathbf{D}^{\mathrm{T}}\mathbf{D}^{\perp} = \mathbf{0}$, Equation (12) can be rewritten as

$$\left(\mathbf{D}^{\perp}\right)^{\mathrm{T}}\mathbf{B}\bar{\mathbf{x}} \equiv \bar{\mathbf{C}}\bar{\mathbf{x}} = \mathbf{0} \tag{13}$$

where $\bar{\mathbf{C}} \equiv \left(\mathbf{D}^{\perp}\right)^{\mathrm{T}}\mathbf{B} \in \mathbb{R}^{q\times m}$. If we can identify the matrix $\bar{\mathbf{C}}$ consistently, the process model is accurately identified up to a similarity transformation. From Equation (13) we know that $\bar{\mathbf{x}}$ is in the orthogonal complement of $\bar{\mathbf{C}}$, i.e.

$$\bar{\mathbf{x}} = \left(\bar{\mathbf{C}}^{\mathrm{T}}\right)^{\perp}\mathbf{s} \tag{14}$$

where $\mathbf{s}$ is an independent vector, which leads to

$$\mathbf{x} = \left(\bar{\mathbf{C}}^{\mathrm{T}}\right)^{\perp}\mathbf{s} + \mathbf{n} \tag{15}$$

Equation (15) has a similar form to the PCA model and is an equivalent representation of Equation (12).

From Equation (15) we observe that the dimension of independently varying factors $\mathbf{s}$ is $m-q$. For a multivariable process under feedback control, these independently varying factors include (i) unmeasured disturbance changes, (ii) measured disturbance changes and (iii) possible setpoint changes.

One note is in order on the persistent excitation of the data used for process monitoring. When system identification models are used for control purposes [52], it is required that the data are persistently excited. In process monitoring, however, if the process data are not persistently excited, i.e. some of the independent factors are not active, the identified PCA model will have fewer principal components than $m-q$. In this case the model can still be used for process monitoring if these inactive factors remain silent. If these factors become active, a process monitoring alarm will be triggered, which indicates a new active mode rather than an actual fault. In this case the process model can be simply updated with the data that reflect the new active mode in order to avoid future false alarms [22].

Another note is on causal models versus correlation-based models. Industrial processes usually operate under highly constrained conditions due to material and energy balances, quality requirements, operational constraints, safety constraints and control feedback. These constrained conditions appear as correlation in the operational data. When PCA and PLS models are built from these data, they are correlation-based models and cannot be interpreted as causal relations. Yoon and MacGregor [53] discuss the difference between causal models and correlation-based models. Care must be taken using correlation models for fault diagnosis. However, if the process data are generated from designed experiments, it is possible to build causal models using PCA if the noise-to-signal ratio is low. Such a model is equivalent to a total least squares approach. If the noise-to-signal ratio is high, direct PCA modeling leads to poor models [24, 53], and system identification techniques [24] or PCA with proper instrumental variables [54] should be used.

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3. FAULT DETECTION INDICES

Fault detection is usually the first step in multivariate process monitoring. Typically the SPE (or Q statistic) and Hotelling's T^2 statistic are used to represent the variability in the RS and PCS respectively. Owing to the complementary nature of these two indices, combined indices are also proposed for fault detection and diagnosis [51, 55]. Another statistic that measures the variability in the RS is Hawkins' statistic [5]. The global Mahalanobis distance test can also be used as a combined measure of variability in the PCS and RS. Individual tests of PCs can also be conducted [5], but they are often not preferred in practice, since one has to monitor many statistics. In this section we summarize these fault detection indices and provide a unified representation.

3.1. Squared prediction error

The SPE index measures the projection of the sample vector on the residual subspace:

$$\text{SPE} \equiv ||\tilde{\mathbf{x}}||^2 = ||(\mathbf{I} - \mathbf{P}\mathbf{P}^{\mathrm{T}})\mathbf{x}||^2 \tag{16}$$

The process is considered normal if

$$\text{SPE} \leq \delta_\alpha^2 \tag{17}$$

where δ_α^2 denotes the upper control limit for SPE with a significance level α. Jackson and Mudholkar [56] developed an expression for δ_α^2:

$$\delta_\alpha^2 = \theta_1 \left(\frac{c_\alpha \sqrt{2\theta_2 h_0^2}}{\theta_1} + 1 + \frac{\theta_2 h_0 (h_0 - 1)}{\theta_1^2} \right)^{1/h_0} \tag{18}$$

where

$$\theta_i = \sum_{j=l+1}^{m} \lambda_j^i, \qquad i = 1, 2, 3 \tag{19}$$

$$h_0 = 1 - \frac{2\theta_1 \theta_3}{3\theta_2^2} \tag{20}$$

l is the number of retained principal components and c_α is the normal deviate corresponding to the upper $1-\alpha$ percentile. Note that this result is derived under the following conditions.

- The sample vector $\mathbf{x}$ follows a multivariate normal distribution.
- An approximation for the distribution is made in deriving the control limit, which is valid when θ_1 is very large.
- The result holds regardless of how many principal components are retained in the model.

When a fault occurs, the faulty sample vector $\mathbf{x}$ is composed of the normal portion superimposed with the fault portion. The fault can make SPE larger than δ_α^2, leading to the detection of the fault.

An alternative upper control limit for SPE is derived in Reference [26] using the result in Reference [57]:

$$\delta_\alpha^2 = g\chi_{h;\alpha}^2 \tag{21}$$

where

$$g = \theta_2/\theta_1, \qquad h = \theta_1^2/\theta_2$$

The relationship between Equation (21) and Equation (18) is discussed in Reference [26].

3.2. Hotelling's T^2 statistic and Hawkins' T_H^2 statistic

Hotelling's T^2 statistic measures variations in the PCS:

$$T^2 = \mathbf{x}^{\mathrm{T}}\mathbf{P}\mathbf{\Lambda}^{-1}\mathbf{P}^{\mathrm{T}}\mathbf{x} \tag{22}$$

Under the condition that the process is normal and the data follow a multivariate normal distribution, the T^2 statistic is related to an F distribution considering that the population mean and covariance are estimated from data [58]:

$$\frac{N(N-l)}{l(N^2-1)} T^2 \sim F_{l,N-l} \tag{23}$$

where $F_{l,N-l}$ is an F distribution with l and $N-l$ degrees of freedom. For a given significance level α the process is considered normal if

$$T^2 \leq T_\alpha^2 \equiv \frac{l(N^2-1)}{N(N-l)} F_{l,N-l;\alpha} \tag{24}$$

If one considers that the mean is accurately known and only the covariance is estimated from data, the T^2 upper control limit is [5]:

$$T_\alpha^2 = \frac{l(N-1)}{N-l} F_{l,N-l;\alpha} \tag{25}$$

Note that the difference between the above two expressions is only a factor of $(N+1)/N$. If the number of data points, N, is so large that the mean and covariance estimated from data are accurate, the T^2 index can be well approximated with a χ^2 distribution with l degrees of freedom and

$$T_\alpha^2 = \chi_{l;\alpha}^2 \tag{26}$$

This result can also be obtained using Reference [57] directly on Equation (22). In process monitoring it is typically true that N is very large. Therefore the χ^2 upper control limit is adequate and often used in the process monitoring literature.

Hawkins' T_{H}^2 statistic is a symmetric implementation of Hotelling's T^2 statistic in the residual subspace [59]:

$$T_{\mathrm{H}}^2 = \mathbf{x}^{\mathrm{T}}\tilde{\mathbf{P}}\tilde{\mathbf{\Lambda}}^{-1}\tilde{\mathbf{P}}^{\mathrm{T}}\mathbf{x} \sim \frac{(m-l)(N^2-1)}{N(N-m+l)} F_{m-l,N-m+l} \tag{27}$$

A drawback of this statistic compared with SPE is that ill-conditioning can result when some of the residual eigenvalues λ_i $(i = l+1, \ldots, m)$ are very close to zero. Similarly, if N is large, the process is considered normal if

$$T_{\mathrm{H}}^2 \leq \chi_{m-l;\alpha}^2 \tag{28}$$

for a significance level α. This test has been used in gross error detection [28] on the residuals due to conservation equations.

3.3. Mahalanobis distance

The Mahalanobis distance, which is defined as follows, forms the global Hotelling's T^2 test:

$$D = \mathbf{x}^{\mathrm{T}}\mathbf{S}^{-1}\mathbf{x} \sim \frac{m(N^2-1)}{N(N-m)} F_{m,N-m} \tag{29}$$

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where $\mathbf{S}$ is the sample covariance of $\mathbf{x}$. When $\mathbf{S}$ is singular with $\text{rank}(\mathbf{S}) = r < m$, Mardia [60] discusses the use of the pseudoinverse of $\mathbf{S}$, leading to a Mahalanobis distance for the reduced rank covariance matrix:

$$D_r = \mathbf{x}^{\mathrm{T}}\mathbf{S}^{+}\mathbf{x} \sim \frac{r(N^2-1)}{N(N-r)}F_{r,N-r} \tag{30}$$

where $\mathbf{S}^{+}$ is the Moore–Penrose pseudoinverse. Mardia [60] further shows that the distance D_r is independent of the type of pseudoinverse. Hence we choose the Moore–Penrose pseudoinverse.

It is straightforward to show that the global Mahalanobis distance is the sum of T^2 in PCS and T^2_{H} in the RS:

$$D = T^2 + T^2_{\mathrm{H}} \tag{31}$$

In process monitoring where the number of observations, N, is rather large, the global Mahalanobis distance approximately follows a χ^2 distribution with m degrees of freedom,

$$D \sim \chi^2_m \tag{32}$$

and the reduced Mahalanobis distance

$$D_r \sim \chi^2_r \tag{33}$$

The upper control limits for D and D_r can be defined accordingly.

3.4. Combined indices

In practice, a single index, rather than two indices, is preferred for monitoring a process. Yue and Qin [14, 51] propose a combined index for fault detection, which combines SPE and T^2 as follows:

$$\varphi = \frac{\text{SPE}(\mathbf{x})}{\delta^2_\alpha} + \frac{T^2(\mathbf{x})}{\chi^2_{l;\alpha}} = \mathbf{x}^{\mathrm{T}}\boldsymbol{\Phi}\mathbf{x} \tag{34}$$

where

$$\boldsymbol{\Phi} = \frac{\mathbf{P}\boldsymbol{\Lambda}^{-1}\mathbf{P}^{\mathrm{T}}}{\chi^2_{l;\alpha}} + \frac{\mathbf{I}-\mathbf{P}\mathbf{P}^{\mathrm{T}}}{\delta^2_\alpha} = \frac{\mathbf{P}\boldsymbol{\Lambda}^{-1}\mathbf{P}^{\mathrm{T}}}{\chi^2_{l;\alpha}} + \frac{\tilde{\mathbf{P}}\tilde{\mathbf{P}}^{\mathrm{T}}}{\delta^2_\alpha} \tag{35}$$

Notice that $\boldsymbol{\Phi}$ is symmetric and positive definite.

In order to apply this index for fault detection, Yue and Qin [51] derive the upper control limit for φ based on the result of Box [57], which provides an approximate distribution having the same first two moments as the exact one. Using the approximate distribution in Reference [57], φ approximately follows

$$\varphi = \mathbf{x}^{\mathrm{T}}\boldsymbol{\Phi}\mathbf{x} \sim g\chi^2_h \tag{36}$$

where the coefficient

$$g = \frac{\text{tr}(\mathbf{S}\boldsymbol{\Phi})^2}{\text{tr}(\mathbf{S}\boldsymbol{\Phi})} \tag{37}$$

and the degree of freedom for the χ^2 distribution is

$$h = \frac{[\text{tr}(\mathbf{S}\boldsymbol{\Phi})]^2}{\text{tr}(\mathbf{S}\boldsymbol{\Phi})^2} \tag{38}$$

It is shown in the Appendix that

$$\text{tr}(\mathbf{S}\boldsymbol{\Phi}) = \frac{l}{\chi^2_{l;\alpha}} + \frac{\sum_{i=l+1}^{m}\lambda_i}{\delta^2_\alpha} \tag{39}$$

and

$$\text{tr}(\mathbf{S}\boldsymbol{\Phi})^2 = \frac{l}{\chi^4_{l;\alpha}} + \frac{\sum_{i=l+1}^{m}\lambda_i^2}{\delta^4_\alpha} \tag{40}$$

After g and h are calculated, the upper control limit of φ can be obtained for a given significance level α; a fault is detected by φ if

$$\varphi > g\chi^2_{h;\alpha} \tag{41}$$

It is worth noting that Raich and Cinar [55] suggest the combined statistic

$$c\frac{\text{SPE}(\mathbf{x})}{\delta^2_\alpha} + (1-c)\frac{T^2(\mathbf{x})}{\chi^2_{l;\alpha}} \tag{42}$$

where $c \in (0, 1)$ is a constant. They further give a rule that the statistic less than one is considered normal. This, however, can lead to erroneous results, because it is possible to have either $\text{SPE}(\mathbf{x}) > \delta^2_\alpha$ or $T^2(\mathbf{x}) > \chi^2_{l;\alpha}$ even if the above statistic is less than one.

3.5. Unified form of the quadratic indices

Tong and Crowe [41] compared several univariate and collective statistical tests, including the global Mahalanobis distance. In this subsection we provide a unified form for these indices available in the literature and compare them in terms of strengths and weaknesses.

Denoting

$$\bar{\mathbf{t}} = [\mathbf{P}\ \ \tilde{\mathbf{P}}]^{\mathrm{T}}\mathbf{x} \tag{43}$$

as the principal components, the above fault detection indices can be unified as follows:

$$\begin{aligned} d &= \bar{\mathbf{t}}^{\mathrm{T}}\mathbf{A}^{+}\bar{\mathbf{t}} \\ &= \bar{\mathbf{t}}^{\mathrm{T}}\text{diag}\{a_1\ a_2 \cdots a_m\}^{+}\bar{\mathbf{t}} \\ &= \sum_{i=1}^{m} a_i^{+}t_i^2 \end{aligned}$$

where

$$\mathbf{A} = \begin{cases} \mathbf{A}_{\text{SPE}} \equiv \text{diag}\{\underbrace{0\cdots0}_{l}\ 1\cdots1\} & \text{for SPE} \\ \mathbf{A}_{T^2_{\mathrm{H}}} \equiv \text{diag}\{\underbrace{0\cdots0}_{l}\ \lambda_{l+1}\cdots\lambda_m\} & \text{for } T^2_{\mathrm{H}} \\ \mathbf{A}_{T^2} \equiv \text{diag}\{\lambda_1\cdots\lambda_l\ 0\cdots0\} & \text{for } T^2 \\ \mathbf{A}_{D_r} \equiv \text{diag}\{\lambda_1\cdots\lambda_r\ 0\cdots0\} & \text{for } D_r \\ \mathbf{A}_D \equiv \boldsymbol{\Lambda} & \text{for } D \\ \mathbf{A}_\varphi \equiv \text{diag}\{\lambda_1\chi^2_{l;\alpha}\cdots\lambda_l\chi^2_{l;\alpha}\ \delta^2_\alpha\cdots\delta^2_\alpha\} & \text{for } \varphi \end{cases}$$

Two major differences among these indices are the scaling of the principal components and the subspace each index covers. SPE, T^2, T^2_{H} and D_r cover a subspace of the entire measurement space. The Mahalanobis distance D, if $\mathbf{S}$ is nonsingular, covers the entire space. The combined index covers the entire space whether $\mathbf{S}$ is singular or not, thus providing a complete measure of the variability in the entire space. Early work by Takemura [61] shows that in practice the Mahalanobis distance often gives too wide upper control limits. The T^2 index in the PCS is better than D, because it avoids the inversion of small but non-zero eigenvalues. The combined index φ is similar to D in that it is a global index, but it avoids the inversion of near-zero eigenvalues.

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These aforementioned indices (SPE and T^2_H in the RS, T^2 in the PCS, the Mahalanobis distance and the combined index) are all collective statistical tests. They are preferable to univariate statistical tests, because the correlation in the data is taken into account. One common feature is that they are all quadratic-form indices. The in-control regions for these indices can be different owing to the possibility of near-zero eigenvalues. The in-control region defined by SPE and T^2 jointly is the joint of two ellipsoids, each in one subspace, while the combined index and the Mahalanobis distance define a global ellipsoid in the measurement space. In the case of process monitoring, SPE is often preferred to T^2, as explained in the next subsection, while in other cases, such as quality control, the T^2 can be preferred. In these cases it is beneficial to use subspace-based indices such as SPE or T^2. If both indices or subspaces are equally important, it is desirable to use a global index such as the combined index φ. This will result in one index to monitor instead of two, and the diagnosis can be done using methods such as contribution plots, which reveals much more information than the two indices.

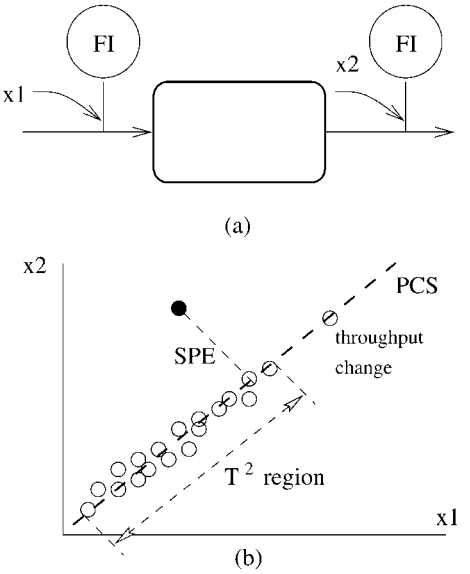


Figure 1. Two flow sensors measuring the inlet and outlet flow of a unit.

3.6. The asymmetric role of SPE and T^2 in process monitoring

Although both SPE and T^2 are used for process monitoring, it is necessary to point out that they measure different situations of the process, and their roles in process monitoring are not symmetric. The SPE index measures variability that breaks the normal process correlation, which often indicates an abnormal situation. The T^2 index measures the distance to the origin in the principal component subspace. Since the principal component subspace typically contains normal process variations with large variance that represent signals, and the residual subspace contains mainly noise, the normal region defined by the control limit for T^2 is usually much larger than that of SPE. Therefore it usually takes a much larger fault magnitude to exceed the T^2 control limit. The normal region defined by the SPE control limit includes residual components that are mainly noise. Therefore faults with small to moderate magnitudes can easily exceed the SPE control limit. Furthermore, if a sample exceeds the T^2 limit only but does not violate the SPE limit, it does not break the correlation structure but simply shifts further away from the origin in the PCS. This case could be a fault, but it could also be a change in the operating region which is not necessarily a fault.

The use of SPE and T^2 for fault detection can be explained with a simple example shown in Figure 1(a), which has two flow sensors measuring the inlet and outlet flow rates of a unit. Under normal steady state operation the data are shown in Figure 1(b). The PCA model with one PC is the 45° line, which is the PCS. A faulty sample (full circle) deviates from the normal model line and increases SPE. This fault breaks the mass balance and is clearly detected using SPE. Note that the T^2 index is inside the control limit in this case. While a fault can cause SPE and T^2 to increase, an increase in T^2 alone indicates that the change is consistent with the model; it may be just a shift of operating region. For example, a change in throughput of the process shown in Figure 1(a) will increase T^2 only, which is not a fault. Therefore we consider the use of SPE for fault detection more preferable than the T^2 index.

Another difference between the PCS and RS is the stationarity of the projected vectors $\hat{\mathbf{x}}$ and $\tilde{\mathbf{x}}$. Real process signals are hardly normally distributed and stationary. While PCA decomposition does not require the signals to be normal or stationary, the principal component subspace usually captures the non-stationary parts of the signals, because they have large variability. Use of the T^2 index can incur false alarms due to the non-stationary, non-normal signals. On the other hand, the T^2 control limit defined by the non-stationary signals can be very wide, which increases the rate of undetected faults. After the major variability is extracted in the PCS, the residual $\tilde{\mathbf{x}}$ usually looks much more stationary and random, which makes the SPE control limit valid. Therefore SPE may have lower chances of type I and type II errors compared with T^2. The T^2 index, however, is more suitable for monitoring quality variables [5], since in-control quality variables are usually stationary.

When the process variables are considered stationary and their number is large, the principal components, which are linear combinations of the variables, can often be approximated with normal distributions owing to the central limit theorem, even though the original variables are not normally distributed. However, it would be dangerous to generalize this statement to considering that *all* major components are normally distributed, because, if so, this would imply that the original variables are normally distributed as they are essentially linear combinations of the major components. If the process variables are highly correlated and the number of independently varying components is not very large, the applicability of the central limit theorem is discounted. One such example is the boiler process data in Reference [10], where one principal component captures the major trend and most of the variance in the data but is obviously not normally distributed.

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4. FAULT ESTIMATION AND RECONSTRUCTION

After a fault is detected, it is important to identify the fault and apply the necessary corrective actions to eliminate the abnormal condition. The procedure to restore normal conditions by applying a corrective change in the data is called data reconstruction, and the procedure for identifying a fault by reconstruction for a given type of faults is called identification via reconstruction. Reconstruction of the normal data from faulty measurements leads to the estimation of the fault magnitude. Therefore fault reconstruction is presented first, followed by fault identification.

The use of PCA models for missing value replacement using simultaneous multiple components is proposed in Reference [37], although the PCA-based method first appeared in Reference [62] using a single-principal-component projection. In the case of missing values the variables that need to be estimated are known. Wise and Ricker [6] apply PCA for faulty sensor detection and reconstruction. Nelson *et al*. [39] discuss the missing value problem using both single-component and multiple-component projections. Dunia *et al*. [10] develop a PCA-based approach for sensor fault detection, reconstruction and identification. It is straightforward to show that the above methods are all equivalent. The reconstruction approach, however, is not limited to sensor reconstruction only; it can be applied to reconstruct the data along an arbitrary direction or subspace [12, 13] provided that there is enough redundancy to reconstruct. In the remaining part of this section we review reconstruction methods for a general fault direction matrix, of which sensor reconstruction is considered a special case.

4.1. Fault reconstruction via optimization

The task of fault reconstruction is to estimate the normal values $\mathbf{x}^*$ by eliminating the effect of a fault $\mathcal{F}_i$. A reconstructed sample vector $\mathbf{x}_i$ is calculated by correcting the effect of a fault on the process data $\mathbf{x}$:

$$\mathbf{x}_i = x - \Xi_i \mathbf{f}_i \tag{44}$$

where $\mathbf{f}_i$ is an estimate of the actual fault magnitude $\mathbf{f}$ along direction Ξ_i.

The objective for reconstruction is to find $\mathbf{f}_i$ such that the reconstructed SPE

$$\mathrm{SPE}(\mathbf{x}_i) = ||\tilde{\mathbf{x}}_i||^2 = ||\tilde{\mathbf{x}} - \Xi_i \mathbf{f}_i||^2 \tag{45}$$

is minimized. The optimal solution to this problem leads to an optimal estimate of the fault magnitude:

$$\mathbf{f}_i = \tilde{\Xi}_i^+ \tilde{\mathbf{x}} = \tilde{\Xi}_i^+ \mathbf{x} \tag{46}$$

The second equal sign is due to the fact that $(\mathbf{I} - \mathbf{C})^2 = \mathbf{I} - \mathbf{C}$. The reconstructed measurement vector is

$$\mathbf{x}_i = \mathbf{x} - \Xi_i \tilde{\Xi}_i^+ \mathbf{x} = \left(\mathbf{I} - \Xi_i \tilde{\Xi}_i^+\right)\mathbf{x} \tag{47}$$

and in the residual space

$$\tilde{\mathbf{x}}_i = \left(\mathbf{I} - \tilde{\Xi}_i \tilde{\Xi}_i^+\right)\tilde{\mathbf{x}} \tag{48}$$

The reconstructed SPE becomes

$$\mathrm{SPE}(\mathbf{x}_i) = ||\tilde{\mathbf{x}}_i||^2 = \tilde{x}^{\mathrm{T}}\left(\mathbf{I} - \tilde{\Xi}_i \tilde{\Xi}_i^+\right)\tilde{\mathbf{x}} \tag{49}$$

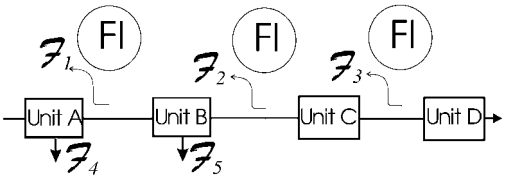


Figure 2. An example that illustrates a set of faults for reconstruction.

The relation $\left(\mathbf{I} - \tilde{\Xi}_i \tilde{\Xi}_i^+\right)^2 = \mathbf{I} - \tilde{\Xi}_i \tilde{\Xi}_i^+$ is used in the above expression. No matter whether it is a sensor fault or a process fault, Dunia and Qin [13] demonstrate that the reconstructed $\mathrm{SPE}(\mathbf{x}_i)$ completely eliminates the impact of the fault, as will be illustrated in the next example.

Example 1

Consider three sensors in the process depicted in Figure 2. The process has four units in series, which operate at steady state. Under normal conditions the three ow sensors give essentially the same readings at any given time, except for measurement noise. Assuming that the standard deviations of the measurement noise for the three sensors are identical, the PCA model will have one principal component with the loading vector

$$\mathbf{P}^{\mathrm{T}} = \frac{\sqrt{3}}{3}[1\ 1\ 1]$$

Now consider that the second sensor fails and is reconstructed from the remaining normal sensors. The corresponding fault direction vector is

$$\Xi_2^{\mathrm{T}} = [0\ 1\ 0]$$

The projection of Ξ_2 on the RS is

$$\tilde{\Xi}_2 = (\mathbf{I} - \mathbf{p}\mathbf{p}^{\mathrm{T}})\Xi_2 = [-1/3\ \ 2/3\ \ 1/3]^{\mathrm{T}}$$

The reconstruction vector according to Equation (47) is

$$\mathbf{x}_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0.5 & 0 & 0.5 \\ 0 & 0 & 1 \end{bmatrix}\mathbf{x} = \begin{bmatrix} x_1 \\ (x_1 + x_3)/2 \\ x_3 \end{bmatrix}$$

Therefore the best reconstruction for sensor 2 is the average of sensors 1 and 3. For an arbitrary process fault direction, say

$$\Xi_i = \frac{1}{\sqrt{3}}[1\ \ 1\ \ -1]^{\mathrm{T}}$$

the reconstruction according to Equation (47) is

$$\mathbf{x}_i = \frac{1}{4}\begin{bmatrix} 3x_1 - x_2 + 2x_3 \\ -x_1 + 3x_2 + 2x_3 \\ x_1 + x_2 + 2x_3 \end{bmatrix} \tag{50}$$

For any given fault magnitude $\mathbf{f}$ in the Ξ_i direction the faulty data vector is

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_1^* \\ x_2^* \\ x_3^* \end{bmatrix} + \frac{1}{\sqrt{3}}\begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}\mathbf{f}$$

The reconstruction using Equation (50) is

$$\mathbf{x}_i = \frac{1}{4}\begin{bmatrix} 3x_1^* - x_2^* + 2x_3^* \\ -x_1^* + 3x_2^* + 2x_3^* \\ x_1^* + x_2^* + 2x_3^* \end{bmatrix}$$

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which completely eliminates the impact of the fault in the reconstruction. This result extends the missing value replacement approach which allows for sensor faults only.

4.2. Related methods

In the case of sensor faults the fault direction matrix Ξ_i takes a special form that corresponds to some columns of an identity matrix. In this special case, other methods such as the missing value replacement method (e.g. References [5, 37, 38]) and the iterative approach of Dunia *et al*. [10] can be applied. Dunia *et al*. [10] further show that these three methods give identical results, unifying the three methods. For a single sensor fault at the ith sensor the fault direction matrix $\Xi_i = [0 \cdots 0,1,0 \cdots 0]^{\mathrm{T}}$, which is the ith column of an identity matrix. The fault estimate from Equation (46) reduces to

$$f_i = x_i - \frac{[\mathbf{c}_{-i}^{\mathrm{T}}\ 0\ \mathbf{c}_{+i}^{\mathrm{T}}]\mathbf{x}}{1 - c_{ii}} \tag{51}$$

where x_i is the ith element of $\mathbf{x}$ and $[\mathbf{c}_{-i}^{\mathrm{T}},\ c_{ii},\ \mathbf{c}_{+i}^{\mathrm{T}}]$ forms the ith row of matrix $\mathbf{C}$, with $\mathbf{c}_{-i}^{\mathrm{T}}$ including the elements before i and $\mathbf{c}_{+i}$ including the elements after i. The relation $(\mathbf{I} - \mathbf{C})^2 = \mathbf{I} - \mathbf{C}$ is used in deriving the above equation. The reconstructed ith variable can be calculated from Equation (47) as

$$x_i^{\mathrm{rec}} = x_i - f_i = \frac{[\mathbf{c}_{-i}^{\mathrm{T}}\ 0\ \mathbf{c}_{+i}^{\mathrm{T}}]\mathbf{x}}{1 - c_{ii}} \tag{52}$$

Although the aforementioned methods give the same results for reconstruction, the optimization method is more general since it works for process faults as well.

Another advantage of the reconstruction method is described by Dunia and Qin [13]: it checks for the reconstructability of a fault and the reliability of the reconstruction. If the fault subspace happens to overlap with the PCS, $\tilde{\Xi}_i$ will be rank-deficient and the fault cannot be completely reconstructed. The pseudoinverse solution in Equation (47) leads to the minimum norm reconstruction. Even though $\tilde{\Xi}_i$ is not rank-deficient, it is possible for the reconstructed values to have a larger variance than the original variables. In this case a better reconstruction or replacement of the faulty sensor is the mean of the variable instead of the reconstruction through the PCA model. This point is further elaborated in Section 7. Furthermore, the variance of the reconstruction error, after reconstructability is guaranteed, can be used to determine the optimal number of principal components [63]. The more traditional cross-validation method for selecting the number of principal components [64] does not guarantee that the left-out values are reconstructable, although a general rule of thumb such as no more than 20% left-out values is recommended in the literature.

4.3. Fault reconstruction using the combined index

In the case where the reconstruction should minimize both SPE and T^2, the combined index of the reconstructed vector can be minimized:

$$\varphi(\mathbf{x}_i) = \frac{\mathrm{SPE}(\mathbf{x}_i)}{\delta_\alpha^2} + \frac{T^2(\mathbf{x}_i)}{\chi_{l;\alpha}^2} = \mathbf{x}_i^{\mathrm{T}}\boldsymbol{\Phi}\mathbf{x}_i \tag{53}$$

Yue and Qin [14, 51] suggest the use of the combined index for reconstruction.

The optimal reconstruction is obtained by minimizing the combined index:

$$\mathbf{f}_i = (\Xi_i^{\mathrm{T}}\boldsymbol{\Phi}\Xi_i)^{-1}\Xi_i^{\mathrm{T}}\boldsymbol{\Phi}\mathbf{x} \tag{54}$$

Yue and Qin [51] demonstrate that $\Xi_i^{\mathrm{T}}\boldsymbol{\Phi}\Xi_i$ is positive definite, which guarantees its invertibility. The reconstructed $\mathbf{x}_i$ and $\varphi(\mathbf{x}_i)$ can be calculated from Equation (54):

$$\mathbf{x}_i = [\mathbf{I} - \Xi_i(\Xi_i^{\mathrm{T}}\boldsymbol{\Phi}\Xi_i)^{-1}\Xi_i^{\mathrm{T}}\boldsymbol{\Phi}]\mathbf{x} \equiv \boldsymbol{\Pi}_{\Xi_i,\boldsymbol{\Phi}}^{\perp}\mathbf{x} \tag{55}$$

where $\boldsymbol{\Pi}_{\Xi_i,\boldsymbol{\Phi}}^{\perp} = \mathbf{I} - \Xi_i(\Xi_i^{\mathrm{T}}\boldsymbol{\Phi}\Xi_i)^{-1}\Xi_i^{\mathrm{T}}\boldsymbol{\Phi}$ is the projection matrix on the orthogonal complement of Ξ_i weighted by $\boldsymbol{\Phi}$, $(\boldsymbol{\Pi}_{\Xi_i,\boldsymbol{\Phi}}^{\perp})^2 = \boldsymbol{\Pi}_{\Xi_i,\boldsymbol{\Phi}}^{\perp}$, and

$$\varphi(\mathbf{x}_i) = x^{\mathrm{T}}\boldsymbol{\Pi}_{\Xi_i,\boldsymbol{\Phi}}^{\perp}\mathbf{x}$$

The use of φ for reconstruction means that the fault is corrected in both the PCS and RS. Owing to the large normal variability in the PCS, it may not always be appropriate to correct in the PCS. Therefore one should assess whether φ should be used for reconstruction as well as for detection.

5. FAULT IDENTIFICATION AND DIAGNOSIS

There has been tremendous interest in diagnosing the possible root causes of a fault situation once it is detected. Thus far the most popular approach to diagnosis is the contribution plot approach [8, 9]. This approach requires no prior knowledge except for a normal PCA or PLS model. The contributions are actually the effects of the fault on the observed vector of measurements. If prior knowledge or historical data of the faults are available, the reconstruction-based approach [10, 12, 13] can lead to more conclusive results. If there are plenty of historical fault records with many fault categories, classification and clustering methods are applicable. In the context of statistical process monitoring, Raich and Cinar [11] apply a similarity index to discriminate among different faults using angles and distances between clusters. Kano *et al*. [65] use a dissimilarity measure to discriminate between the normal and faulty clusters. By using Fisher discriminant analysis, Chiang *et al*. [66] achieve maximum separation between the normal and faulty clusters. This approach is more suitable for detecting mean changes than covariance changes.

In the rest of this section we focus on the reconstruction-based approach and the contribution-based approach. Both approaches are applicable to sensor and process fault diagnosis, but they have different characteristics that are often complementary. The contribution plot approach does not require any information about the fault to generate the plots; in the case where prior knowledge is available, it is up to the user to interpret the plots. The reconstruction-based approach, on the other hand, requires knowledge of the fault directions. In the case where faulty data are available for a process fault, it is straightforward to model the fault directions from the faulty data and use them for future fault identification [50,51]. In this case, knowledge from historical faulty data is built into the method. Owing to the use of

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additional information, the reconstruction-based approach can be more conclusive in the results. If a fault has never happened before but is detected for the first time, the reconstruction-based approach can also extract the fault direction from the faulty data. The interpretation of the fault direction can be done similarly to that of the contribution plots. Further, the fault direction can be used for future fault identification via reconstruction. The fault direction extraction will be discussed in Section 6. An additional benefit of the reconstruction-based approach is that it provides a framework to analyze fault detectability, reconstructability and identifiability, which will be reviewed in Section 7.

5.1. Reconstruction-based approach

The reconstruction approach for fault identification consists of finding the true fault from a set of candidate faults. Dunia and Qin [13] propose a fault identification method by assuming each of the faults in $\{\mathcal{F}_j\}$ in turn and performing reconstruction. When Ξ_j is assumed, which may or may not be the true fault, the reconstructed sample vector is $\mathbf{x}_j = \mathbf{x}_{\Xi} j\mathbf{f}_j$, and $\mathbf{f}_j$ is estimated such that

$$\mathrm{SPE}(\mathbf{x}_j) \equiv ||\tilde{\mathbf{x}}_j||^2 = ||\tilde{\mathbf{x}} - \tilde{\Xi}_j \mathbf{f}_j||^2 \tag{56}$$

is minimized. The least squares solution for $\mathbf{f}_j$ is

$$\mathbf{f}_j = \tilde{\Xi}_j^+ \tilde{\mathbf{x}} \tag{57}$$

The reconstructed vector $\tilde{\mathbf{x}}_j$ can be related to the fault-free vector $\tilde{\mathbf{x}}^*$ as

$$\begin{aligned} \tilde{\mathbf{x}}_j &= \tilde{\mathbf{x}} - \tilde{\Xi}_j \mathbf{f}_j = (\mathbf{I} - \tilde{\Xi}_j \tilde{\Xi}_j^+)\tilde{\mathbf{x}} \\ &= (\mathbf{I} - \tilde{\Xi}_j \tilde{\Xi}_j^+)\tilde{\mathbf{x}}^* + (\mathbf{I} - \tilde{\Xi}_j \tilde{\Xi}_j^+)\tilde{\Xi}_i \mathbf{f} \end{aligned} \tag{58}$$

When the actual fault Ξ_i is assumed, i.e. $j = i$, the second term in Equation (58) is zero, which leads to

$$\tilde{\mathbf{x}}_i = (\mathbf{I} - \tilde{\Xi}_i \tilde{\Xi}_i^+)\tilde{\mathbf{x}}^* \tag{59}$$

Since $\mathbf{I} - \tilde{\Xi}_i \tilde{\Xi}_i^+$ is a projection matrix,

$$||(\mathbf{I} - \tilde{\Xi}_i \tilde{\Xi}_i^+)\tilde{\mathbf{x}}^*|| \leq ||\tilde{\mathbf{x}}^*|| \tag{60}$$

The SPE of the reconstructed vector is

$$\begin{aligned} \mathrm{SPE}(\mathbf{x}_i) = ||\tilde{\mathbf{x}}_i||^2 &= ||(\mathbf{I} - \tilde{\Xi}_i \tilde{\Xi}_i^+)\tilde{\mathbf{x}}^*||^2 \\ &\leq ||\tilde{\mathbf{x}}^*||^2 = \mathrm{SPE}(\mathbf{x}^*) \leq \delta_\alpha^2 \end{aligned} \tag{61}$$

Therefore, when the true fault is assumed, the reconstructed SPE is brought into the control limit. If $\tilde{\Xi}_j \neq \tilde{\Xi}_i$, the last term in Equation (58) is not zero, which makes $\mathrm{SPE}(\mathbf{x}_j)$ outside the SPE control limit given a large enough fault magnitude. The issue of fault identifiability between Ξ_i and Ξ_j is discussed by Dunia and Qin [13]. Section 7 will discuss briey the fault identifiability issue for the case of unidimensional faults. In summary, if $\mathrm{SPE}(\mathbf{x}_j) \leq \delta_\alpha^2$, Ξ_j is considered as the fault; if several Ξ_j are identified, then the true fault is not uniquely identifiable but is identified to a subset.

Yue and Qin [51] identify faults based on the combined index. If the reconstruction in a fault subspace leads to a feasible solution in the normal region, the fault subspace is considered as the true fault. The reconstruction method minimizes φ along Ξ_j in this case. This method is appropriate when both SPE and T^2 are important indices for fault detection and diagnosis. This method, however, requires knowing $\Xi_j = \tilde{\Xi}_j + \tilde{\Xi}_j$, while the SPE-based method requires only $\tilde{\Xi}_j$. If the process fault directions are modeled from data, as shown in References [50, 51], the estimate of $\tilde{\Xi}_j$ is usually less accurate than that of $\tilde{\Xi}_j$ owing to the large normal variability in the PCS, leading to a less accurate estimate for Ξ_j.

5.2. Contribution plots

Contribution plots are well known diagnostic tools for fault identification [8, 9, 40, 43, 67]. The commercial use of contribution plots was patented by Hopkins *et al*. [68] in 1995. The most common indices used for fault diagnosis with contribution plots are SPE and T^2. Contribution plots on SPE indicate the significance of the effect of each variable on the index at different sampling times. If a sample vector $\mathbf{x}$ has an abnormal SPE, the variables that appear to have a significant contribution are investigated. A contribution plot on PCA scores indicates the significance of the effect of each variable on the T^2 index. The variables with the largest contribution are considered major contributors to the fault.

The contribution for SPE is simply breaking down the summation of SPE (Equation (16)) into each element:

$$\mathrm{SPE} = \sum_{i=1}^{m} \tilde{x}_i^2 = \sum_{i=1}^{m} \mathrm{SPE}_i \tag{62}$$

where $\mathrm{SPE}_i = \tilde{x}_i^2$ is the contribution of the ith variable. The T^2 distribution is not as clearly defined owing to the definition of T^2. Miller *et al*. [8] define the contribution for each PC and each variable, which is difficult to use in practice. Nomikos [69] defines a T^2 distribution that involves cross-talk among variables, which could lead to negative contributions. Qin *et al*. [70] define a T^2 contribution that eliminates the cross-talk among variables. Westerhuis *et al*. [71] propose other generalizations to the T^2 contributions by including all principal components. Upper control limits for contribution plots are discussed in References [70–72].

The contribution plots are very easy to calculate, with no prior knowledge required to generate the plots. Prior knowledge, however, is often used and required to interpret the plots. As explained by Kourti and MacGregor [40], the contribution plots may not explicitly identify the cause of an abnormal condition, but they determine the entries in $\mathbf{x}$ that are not consistent with the normal operating conditions. The reason is that the contribution from one variable is propagated to other variables in calculating the projection $\tilde{\mathbf{x}}$. This 'smearing' effect can reduce the difference between contributing and non-contributing variables, which in the extreme case can lead to mis-identification. The reconstruction-based approach with known fault directions completely eliminates the fault when the actual fault direction is used for reconstruction. In the case of arbitrary process fault directions the reconstruction-based approach completely removes the effect of the fault and brings SPE within the normal control limit.

Owing to limited redundancy or correlation among the process variables, it is possible that some faults may not be identifiable. In this case one should be cautious in drawing conclusions from any diagnosis methods. For example, for

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the case of two variables and one PC the loading matrices can be parametrized as

$$\mathbf{P} = \begin{bmatrix} \sin\theta \\ -\cos\theta \end{bmatrix}$$
$$\tilde{\mathbf{P}} = \begin{bmatrix} \cos\theta \\ \sin\theta \end{bmatrix}$$

Consider a fault in sensor 1. The reconstruction-based identification results along sensor 1 and sensor 2 directions are

$$\mathrm{SPE}(\mathbf{x}_1) = 0 < \delta_\alpha^2$$
$$\mathrm{SPE}(\mathbf{x}_2) = 0 < \delta_\alpha^2$$

which indicates explicitly that one of the two is faulty, but they are not identifiable further owing to lack of redundancy. Using the contribution plot approach, the SPE contributions are

$$\mathrm{SPE}_1 = \tilde{x}_1^2 = \cos^2\theta(x_1\cos\theta + x_2\sin\theta)^2$$
$$\mathrm{SPE}_2 = \tilde{x}_2^2 = \sin^2\theta(x_1\cos\theta + x_2\sin\theta)^2$$

Therefore $\mathrm{SPE}_2/\mathrm{SPE}_1 = \tan^2\theta$ regardless of the fault in sensor 1. If the two variables are scaled to unit variance, the angle θ will be 45° and the contributions SPE_1 and SPE_2 are about the same. However, if variable 2 is more important and is given more weight, the SPE_2 contribution will always be larger, which leads to mis-identification. The correct answer to this problem is that there is not enough redundancy to identify the two faults; the identifiability issue will be discussed in Section 7. A similar example is illustrated in Reference [73].

It might also be noted that the contribution plot approach is dependent on the scaling of variables. To demonstrate the effect of scaling on the contribution- and reconstruction-based approaches, we give the following example.

Example 2. Sensor faults

The data generating equation for this example is

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0.3873 & 0.1190 \\ -0.1291 & 0.2379 \\ 0.9037 & -0.1530 \\ 0.1291 & 0.9518 \end{bmatrix} \begin{bmatrix} s_1 \\ s_2 \end{bmatrix} + \text{noise}$$

where s_1 and s_2 are zero-mean random sequences with standard deviations of 1 and 0.8 respectively. We generate 100 data samples to build a PCA model and then generate an additional 12 samples with a bias fault in sensor 3. The noise standard deviation is 0.2 with normal distribution. The standard deviation of the normal data is $[0.47\ 0.34\ 0.99\ 0.85]^{\mathrm{T}}$. We perform fault diagnosis using reconstruction-based and contribution plot approaches. We also consider the case where it is required not to scale the variables owing to other reasons, and the case where the variables are scaled to unit variance.

The fault identification results using the reconstruction-based approach and the contribution plots are shown in Figure 3. When the variables are not scaled to unit variance, the reconstruction-based approach correctly identifies that

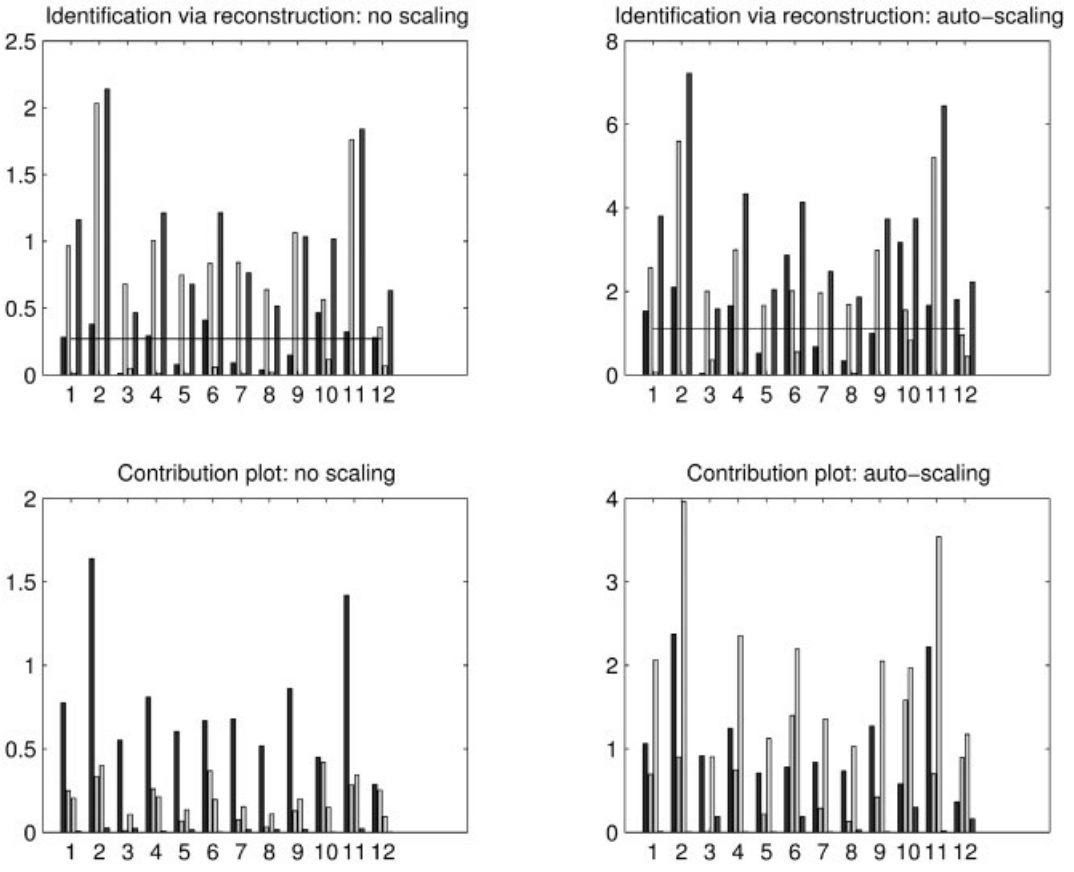


Figure 3. Sensor fault identification results for Example 2. Top plots: each group has four bars representing reconstructed SPE along four sensor directions; smallest value indicates a fault. Bottom plots: each group has four bars representing contributions of four sensors; largest value indicates a fault.

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sensor 3 is faulty, except for the third sample (lowest bar in a group). The contribution plot approach points to sensor 1 being faulty for all samples. When the variables are auto-scaled, as shown in the right two plots of Figure 3, both the reconstruction approach and the contribution plot approach correctly identify sensor 3 as being faulty except for sample 3. This example shows that the contribution plot approach is very sensitive to scaling, while the reconstruction-based approach is not. When the variables are not scaled to unit variance, one should be careful about interpreting the contribution plots. The scaling aspect of contribution plots deserves further study.

5.3. Discussion

To facilitate discussion, we first give the following illustrative example.

Example 3. Process faults

Consider three units in series and four flow sensors as shown in Figure 4. From historical operation it is known that unit B often leaks. Consider five candidate faults: each sensor fault and a leak in unit B. 20% measurement noise is added to the sensors. We create a leak (constant flow) in unit B while the main flow varies randomly. The following tasks are performed:

- calculating reconstruction indices along five directions;
- calculating contributions for each variable.

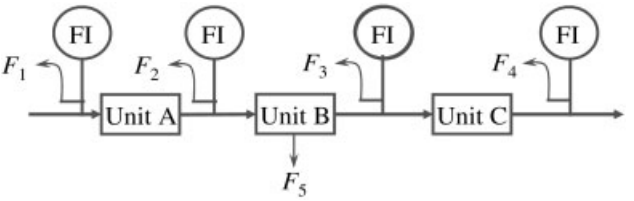


Figure 4. Process and fault diagram for Example 3.

The results are shown in the top and middle plots of Figure 5. It appears that the reconstruction-based approach uniquely identifies the leak in unit B as the true fault. The contribution result in the middle plot does not show any particularly large contributions, hence it is not informative about the fault. If we calculate the SPE contribution in Equation (62) by plotting $\tilde{x}_i$, i.e. without squaring them, the result is as shown in the bottom plot of Figure 5. We observe a consistent pattern in which the first two contributions are positive and the last two contributions are negative. By examining the process diagram, we can conclude that there must be a material loss between sensor 2 and sensor 3, which indicates a leak in unit B.

Example 3 demonstrates the complementary nature of the reconstruction- and contribution-based approaches. If prior knowledge is available to characterize the process fault direction, the reconstruction-based approach can point to the actual cause of the fault. This approach, however, cannot enumerate all possible faults, but simply focuses on typical

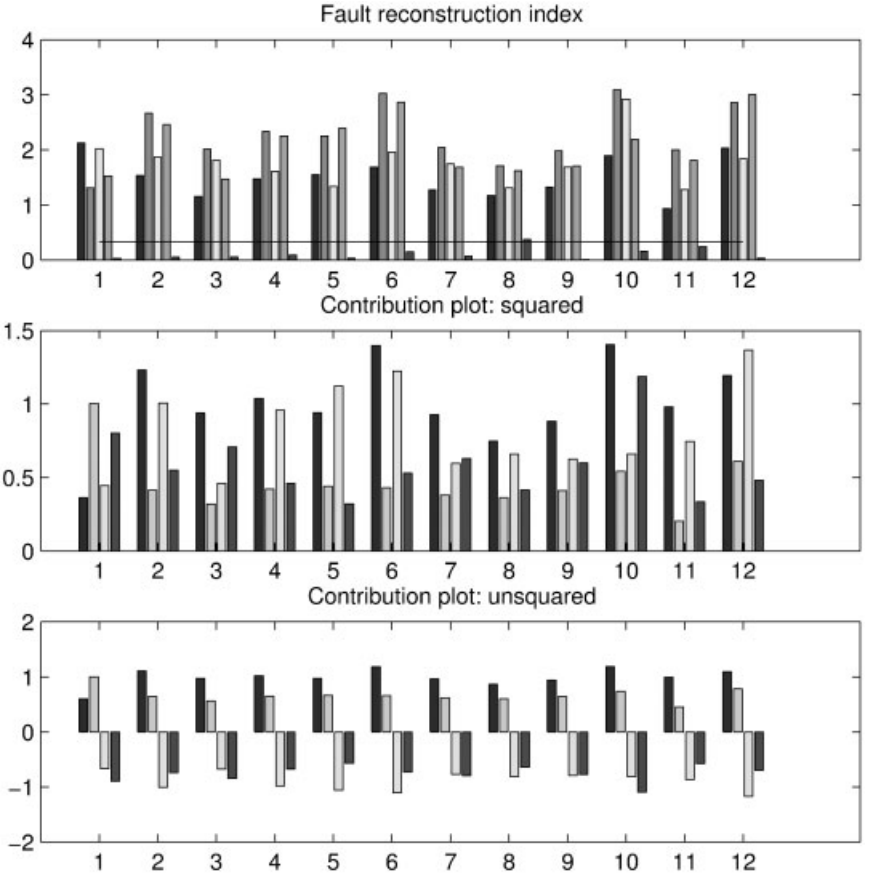


Figure 5. Identification via reconstruction vs contribution plots for Example 3. Top plot: each group has five bars representing reconstructed SPE along four sensor fault and one process fault directions; smallest value indicates the true fault. Middle and bottom plots: each group has four bars representing contributions of four variables.

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faults that are important to the safety and health of the process. The contribution plot approach does not require prior knowledge to calculate the contributions, but requires process knowledge to interpret the contributions.

The use of contribution plots is not always to find the largest contributions. In Example 3 the sign of the contributions is much more informative. For a more complex, realistic process it is up to the user's experience to determine whether one should look at the magnitude or the sign of the contributions, or a combination of both sign and magnitude. For large processes with possibly hundreds of variables it can be overwhelming to examine either the magnitude or the sign of the magnitude of the contributions. Multiblock approaches in this case can be much more informative [9,70]. In the next section the contribution plot approach is discussed in a hierarchical framework with an industrial example. Fault directions extracted from fault data are interpreted similarly to contribution plots and are used for fault identification via reconstruction.

So far we have discussed mainly SPE contributions. The T^2 contributions can also be useful, but the contributions to T^2 break down to individual principal components. The lumped T^2 contributions for each variable have to incur some approximation [69–71]. The reconstruction-based approach can also be done based on T^2 or the combined index of References [51, 74].

In the case where no prior knowledge is available about the fault directions, the reconstruction-based approach can be implemented similarly to the contribution plot approach. In this approach, one simply assumes that each variable is a potential contributor and reconstructs along each variable to minimize the distance to the normal model. The distance measure can be SPE or Mahalanobis distance. The largest correction indicates a large contribution to the fault situation. This approach is suggested by Runger *et al*. [73]. Similarly to the contribution plot approach, this approach is also sensitive to variance scaling. A major advantage of this approach over the contribution plot approach is that the calculation of T^2 contributions does not require approximations as done in References [69–71]. Process knowledge is also needed to interpret these reconstruction-based contributions. The reconstruction-based contributions can also be calculated for the combined index or any quadratic indices.

6. FAULT DIAGNOSIS OF AN INDUSTRIAL FILM PROCESS

This case study shows how the reconstruction-based approach and the hierarchical contribution plot approach can be used for complex process fault diagnosis. The process is a polyester film manufacturing process which is studied in Reference [70]. The raw material, a polyethylene polymer that comes from batch reactors, is first extruded in a chill roll drum to form a film. The film is then biaxially oriented and stretched first in the machine direction and then in the transverse direction. The orientation is accomplished by passing the film over rollers that run at increasingly faster speed ($300\,\mathrm{m\,min^{-1}}$), then fed into a tenter oven, where it is pulled at right angles (transverse direction orientation). This stretching rearranges the polymer molecules into an orderly structure to substantially improve the film's mechanical properties. Finally, the film is heat-set to stabilize it. A number of faults can occur in the process. A typical fault is a sudden oscillation in some temperature loops by about 10°C, which severely affects the quality of the film product. It is desirable to diagnose this fault as soon as it happens.

In the polyester film process there were initially a total of 308 variables, which were reduced to 103 variables with the help of the plant engineers. The variables in the total data set include process variables, setpoints, output variables and monitoring variables. For this analysis, process variables and monitoring variables were used. After a preliminary analysis it was clear that the first 1167 samples can be used to generate the normal PCA model. The period from sample 1168 to sample 1417 contains the typical fault which can be used to test different fault diagnosis methods. For the normal PCA model the number of principal components is determined to be 15 using the variance of the reconstruction error [63].

Figure 6 shows an example of contribution plots for the polyester film process for the faulty samples. The SPE in the top-left plot shows significant violation of the control limit. The bottom-right plot shows a typical contribution plot. For this faulty period the highest contributing variable is shown in the top-right plot, which frequently points to variable 28, variable 25 and sometimes variable 32. The faulty data for variable 28 and its PCA model projection are shown in the bottom-left plot, which shows a large difference between them. This indicates that the contribution plot points to the contributing variable correctly.

Owing to the large number of variables in this example, it can be difficult to interpret the contribution plot when there are many competing large contributors, as in the case of the bottom-right plot in Figure 6. In the next subsection we discuss the use of the reconstruction-based approach for fault diagnosis. The faulty data are divided into two parts. The first part is used to extract the fault direction matrix Ξ_i to characterize this fault. The fault direction matrix is then used to identify major contributing variables. The second part of the faulty data is treated as new faulty data, and the fault direction matrix extracted earlier is used to identify that the second part is essentially the same fault as the first part.

6.1. Fault subspace extraction

When a process is under faulty condition, its measurement will contain the normal values of the process variables and the fault. The kth sample under fault $\mathcal{F}_i$ can be projected to the residual subspace:

$$\tilde{\mathbf{x}}(k) = \tilde{\mathbf{x}}^*(k) + \tilde{\Xi}_i \mathbf{f}(k)$$

It is often the case that the faulty residual is much larger than the normal part projected on the residual subspace, i.e.

$$||\tilde{\mathbf{x}}(k)|| \gg ||\tilde{\mathbf{x}}^*(k)||$$

Then we have

$$\tilde{\mathbf{x}}(k) \approx \tilde{\Xi}_i \mathbf{f}(k)$$

Collecting p observations under fault $\mathcal{F}_i$ and denoting

$$\mathbf{X}_i = [\mathbf{x}(1) \;\; \mathbf{x}(2) \cdots \mathbf{x}(p)]^{\mathrm{T}}$$

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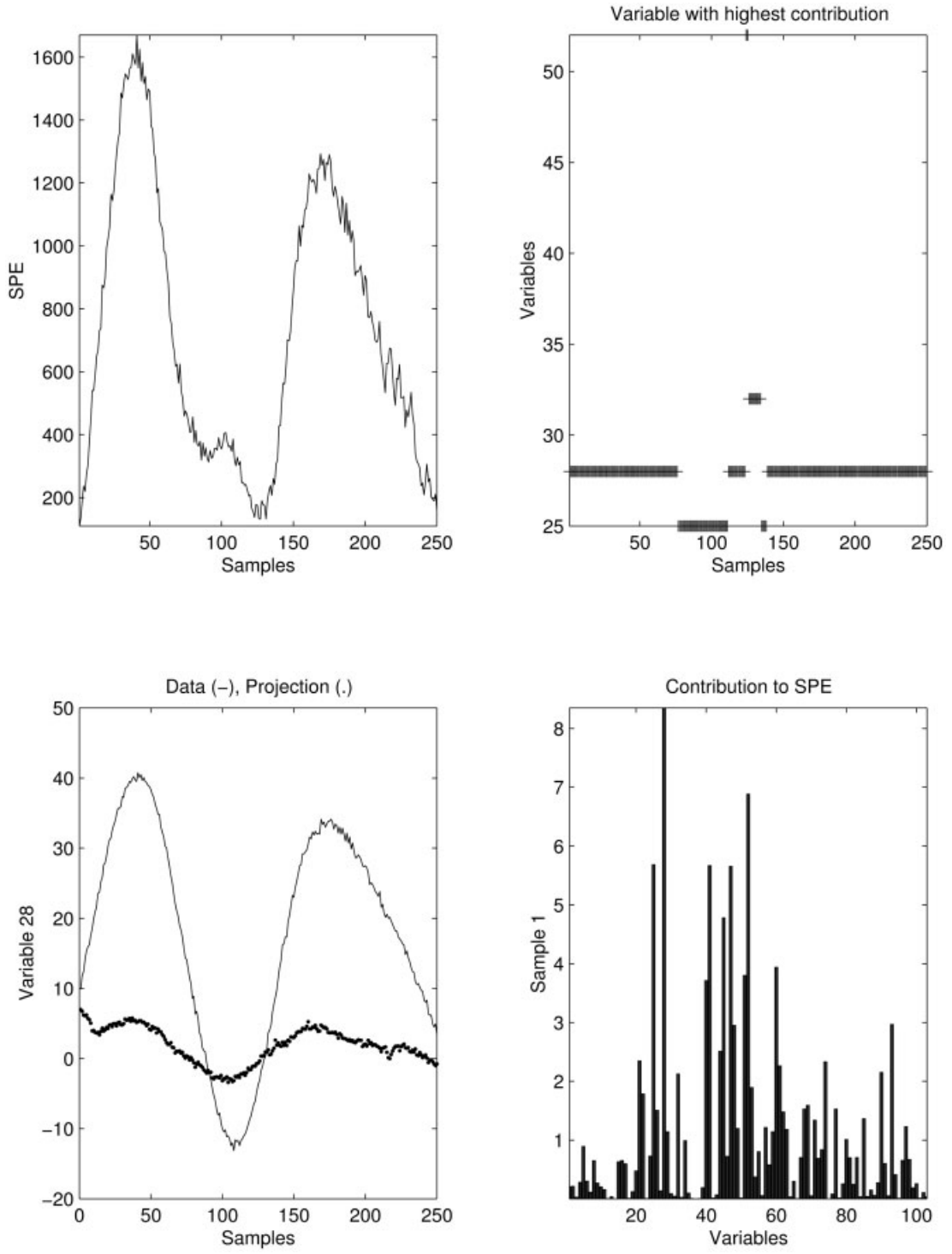


Figure 6. Fault detection and diagnosis results using contribution plots.

we obtain

$$\tilde{\mathbf{X}}_i^{\mathrm{T}} = \tilde{\Xi}_i[\mathbf{f}(1) \quad \mathbf{f}(2) \cdots \mathbf{f}(p)]$$

Therefore $\tilde{\Xi}_i$ shares the same subspace as $\tilde{\mathbf{X}}_i^{\mathrm{T}}$. We apply SVD on the residual matrix $\tilde{\mathbf{X}}_i^{\mathrm{T}}$:

$$\tilde{\mathbf{X}}_i^{\mathrm{T}} = \mathbf{U}_i \mathbf{D}_i \mathbf{V}_i^{\mathrm{T}}$$

The fault direction matrix can be chosen as

$$\tilde{\Xi}_i = \mathbf{U}_i \tag{63}$$

Equation (63) allows us to extract fault subspaces from faulty data. If the faulty data for a number of faults are available, we can extract the corresponding fault directions or subspaces $\{\tilde{\Xi}_i\}$. If one of these faults occurs again in the future, it can be identified via reconstruction. If the existing set of fault subspaces cannot reconstruct a newly detected fault, a new fault direction can then be extracted and stored for future identification.

Whether or not a fault has happened before, the extracted fault directions $\tilde{\Xi}_i$ from faulty data can be interpreted similarly to contribution plots. This feature of the fault

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directions is demonstrated in the next subsection. In the limiting case where only one fault observation is used in fault direction extraction, the fault direction in Equation (63) reduces to the standard contributions to SPE. For a multi-dimensional fault the number of observations required for extracting the fault direction should be larger than the dimension of the fault; more observations often lead to better estimates of the fault directions. The SVD approach to extracting fault directions is not simply averaging the fault contributions over multiple observations. It extracts the common, significant variations due to the effect of the fault, similarly to applying PCA on the faulty residuals.

With the knowledge of $\tilde{\Xi}_i$, Equation (56) can be calculated. Now that we know $\mathrm{SPE}(\mathbf{x}_i)$ and $\mathrm{SPE}(\mathbf{x})$, we can define a fault identification index (FII)

$$\eta_i = \frac{\mathrm{SPE}(\mathbf{x}_i)}{\mathrm{SPE}(\mathbf{x})}$$

whose values range from 0 to 1. If η_i is close to one, Ξ_i is not likely the true fault, since it offers little correction in SPE. If η_i is close to zero, the fault has been identified. The FII works well for process fault identification even though the fault direction estimation is not very accurate.

6.2. Reconstruction-based approach

By applying fault direction extraction to the polyester film manufacturing process, we obtain in Figure 7 the directions extracted from the faulty data. These extractions are made using the data from samples 1–125 in Figure 6. Observe in Figure 7 that variable 28 in dimension 1 and variable 25 in Dimension 2 dominate these directions. Variable 32 in dimension 3 is also significant and variable 40 dominates dimension 4. This result is essentially consistent with the contribution plot results in Figure 6, with the additional suspect, variable 40, being identified. Therefore it is possible to identify variables that are causing the upset in the process using fault direction information.

To adequately extract all dimensions of the fault, the reconstructed SPE after extracting the fault directions should be deated within the normal SPE limit. Figure 8 shows that the SPE is deated every time a dimension is extracted from the faulty data. After extracting dimension 1, the major hump is deflated. After extracting dimension 2, the second major hump is deflated. More than 95% of the reconstructed SPEs fall in the control limit with five dimensions, and all reconstructed SPEs fall in the control limit after the eighth dimension.

After the extraction has been done for a known fault, we can use this fault signature to identify that fault in new faulty data. The top plot in Figure 9 shows the SPE for the new data from samples 126–250. When the extracted directions are applied to this faulty section, we observe in the middle plot of Figure 9 that the reconstructed SPE is deflated. The fault identification index, which is the ratio of the reconstructed SPE to the original SPE, is shown in the bottom plot of Figure 9. This result shows that the same fault is identified up to sample 90, because the reconstructed SPE is small and the FII values are close to zero. After sample 90 the reconstructed SPE cannot be adequately deflated, which indicates that a new disturbance enters the process.

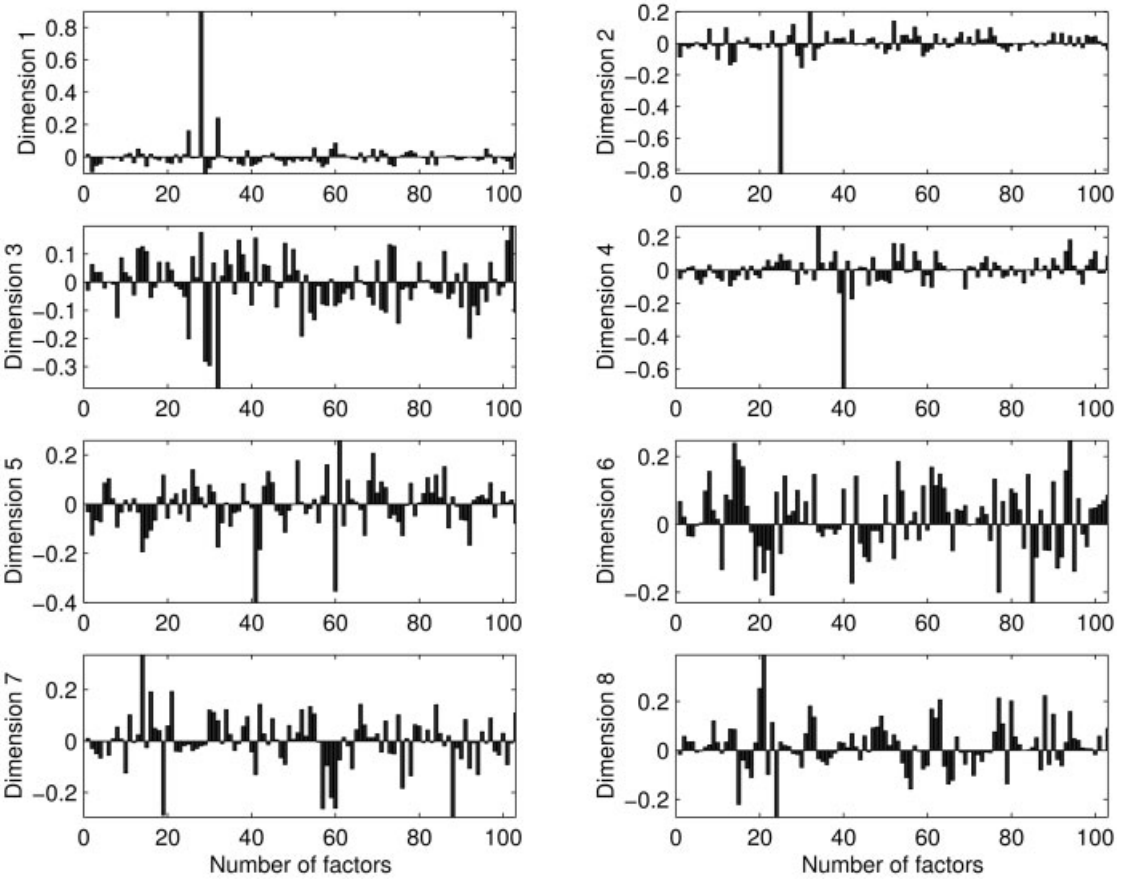


Figure 7. Fault direction in eight dimensions.

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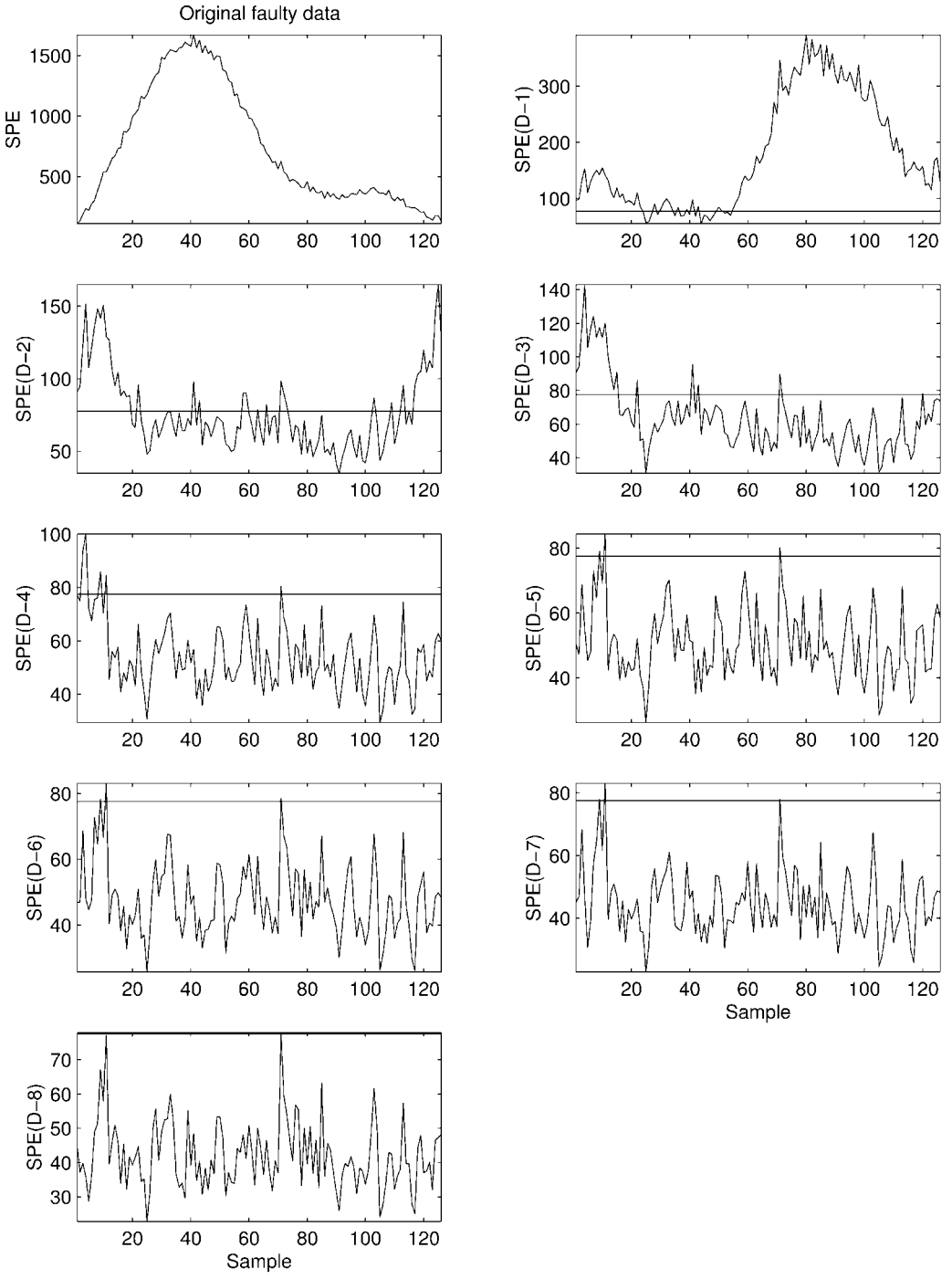


Figure 8. SPEs after extracting each fault dimension.

In summary, when a fault is detected, the faulty data can be used to extract the fault direction matrix. This fault direction matrix can be used to identify major contributing variables to the fault situation, similarly to the contribution plot approach. Furthermore, the directional knowledge of the fault can be used to identify newly detected faults. If a newly detected fault is reconstructed adequately using a known fault direction, one can conclude that the same fault happens again.

6.3. Hierarchical contribution plots

In the case of a large number of variables it can be difficult to interpret the contribution plots. An effective approach is to use multiple-block fault diagnosis [9]. Qin *et al*. [70] report the use of hierarchical contribution plots for this industrial process.

The process is partitioned into seven blocks as shown in Table I. The partitioning is based on the knowledge of the process in terms of process sections. Observe that the sizes of the blocks are very different. It is suggested to divide the process into sections that describe a unit or a specific physical or chemical operation. After grouping into blocks, the overall SPE and T^2 are used to detect a fault. Once a fault is detected, the block SPE and T^2 are calculated and examined against their control limit [70]. If a block SPE or T^2 is outside the control limit, variable contributions in that block are further examined.

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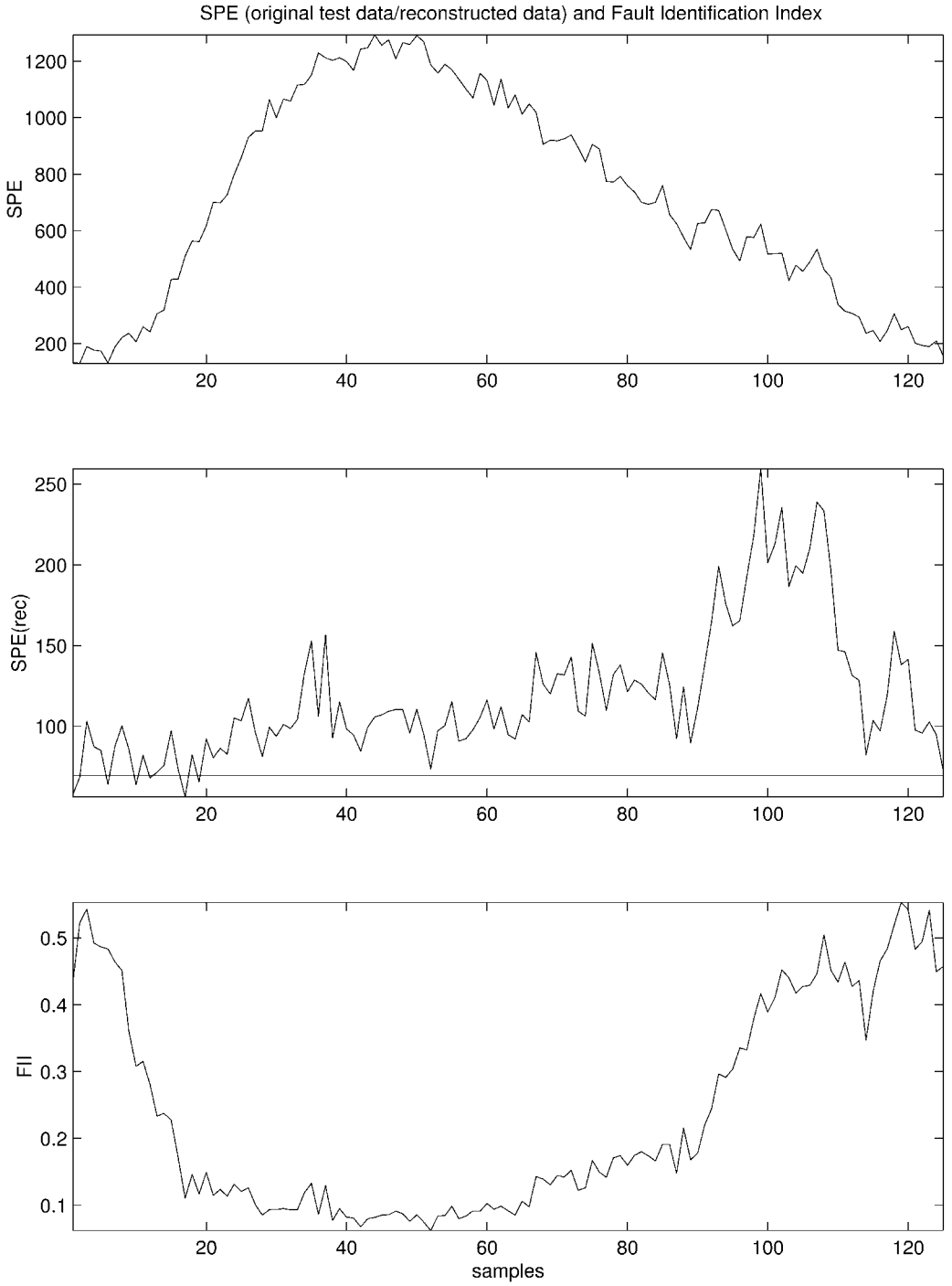


Figure 9. Fault identification results on new faulty data.

Table I. Polyester film manufacturing process variables divided into seven blocks

Block number	Process section	Variables in each block
1	Drying zone	1–9
2	Extrusion zone	10–29
3	Melt pipes zone 1	30–40
4	Melt pipes zone 2	41–52
5	Die zone	53–61
6	Casting zone	62–77
7	Tenter zone	78–103

To identify the fault, a hierarchical contribution plot for SPE is shown in Figure 10. The top plot shows the block contributions. It is found from the top plot that blocks 2 and 3 are the blocks where the fault is located. The bottom plot in Figure 10 shows the contribution plot for SPE for block 2, which clearly identifies that variables 25 and 28 are contributing to the out-of-control situation. Compared with the standard contribution plot result shown in Figure 6, the hierarchical monitoring approach gives a much clearer indication of the faulty variables.

In Figure 11 the top plot shows the contributions to T^2 for each block. Again it is shown clearly that block 2 is

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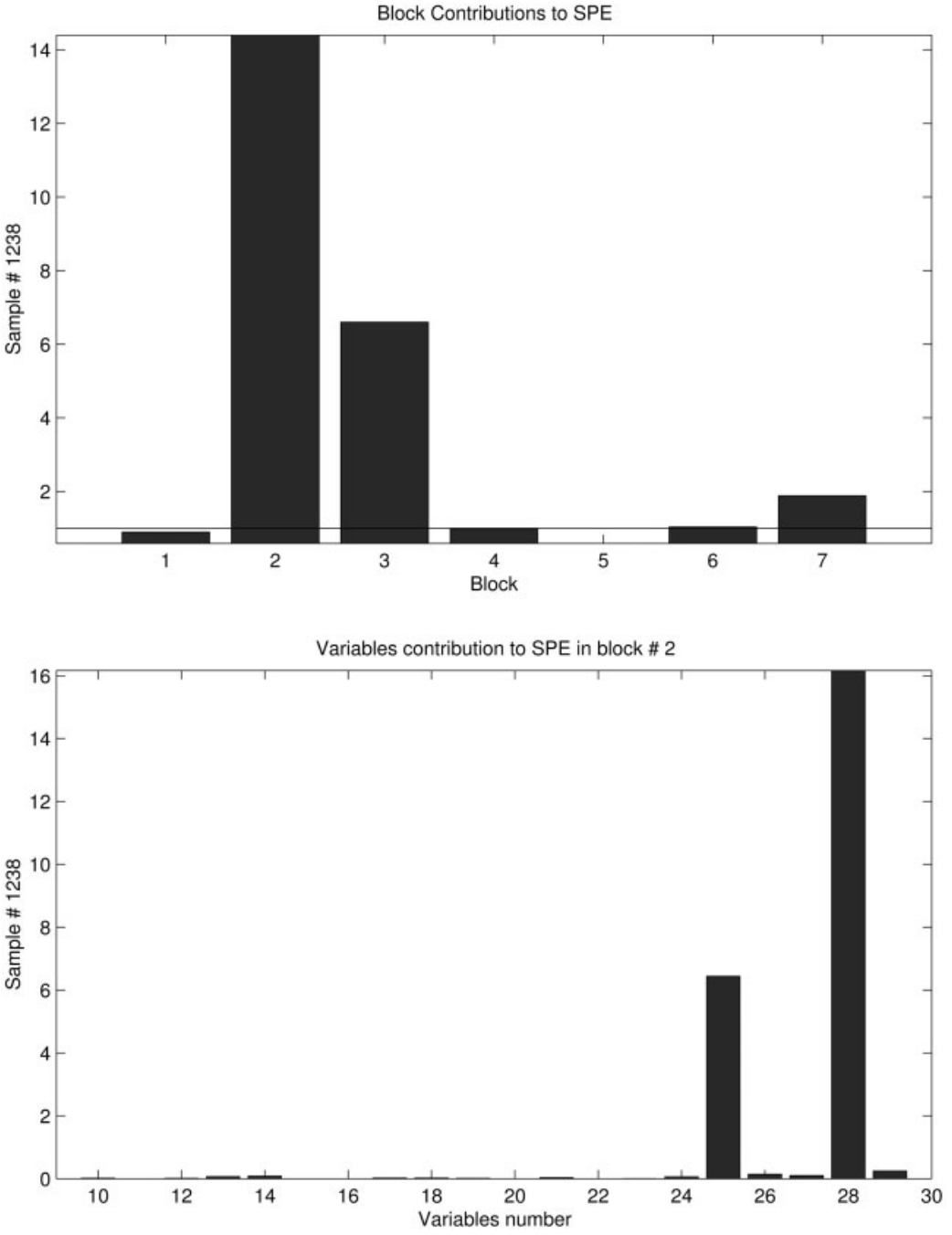


Figure 10. Identification of faulty blocks and contributing variables to SPE.

responsible for the abnormal situation. The bottom plot of Figure 11 shows the contribution to T^2 for Block 2, which again identifies variables 25 and 28 as the major contributors.

The industrial film process clearly demonstrates the complementary nature and duality of the reconstruction-based approach and contribution-based approach to fault diagnosis. The contribution plot approach requires no prior knowledge to calculate the contributions, but needs prior process knowledge to interpret the results. The reconstruction-based approach can be used in two ways. One way is to use it to extract fault directions and interpret the directions similarly to contribution plots. In this way it does not require prior knowledge up front. Another way is to use the extracted fault directions to identify future faults. For important, frequent process faults this approach can be more efficient and conclusive than other approaches, since it uses fault directional knowledge explicitly. The hierarchical monitoring framework shown in this section can be extended to the reconstruction-based approach as well, which leads to even clearer identification results [75].

7. FAULT DETECTABILITY, RECONSTRUCTABILITY AND IDENTIFIABILITY

From both an analytical and a practical point of view it is important to know whether a fault is detectable, reconstruct-

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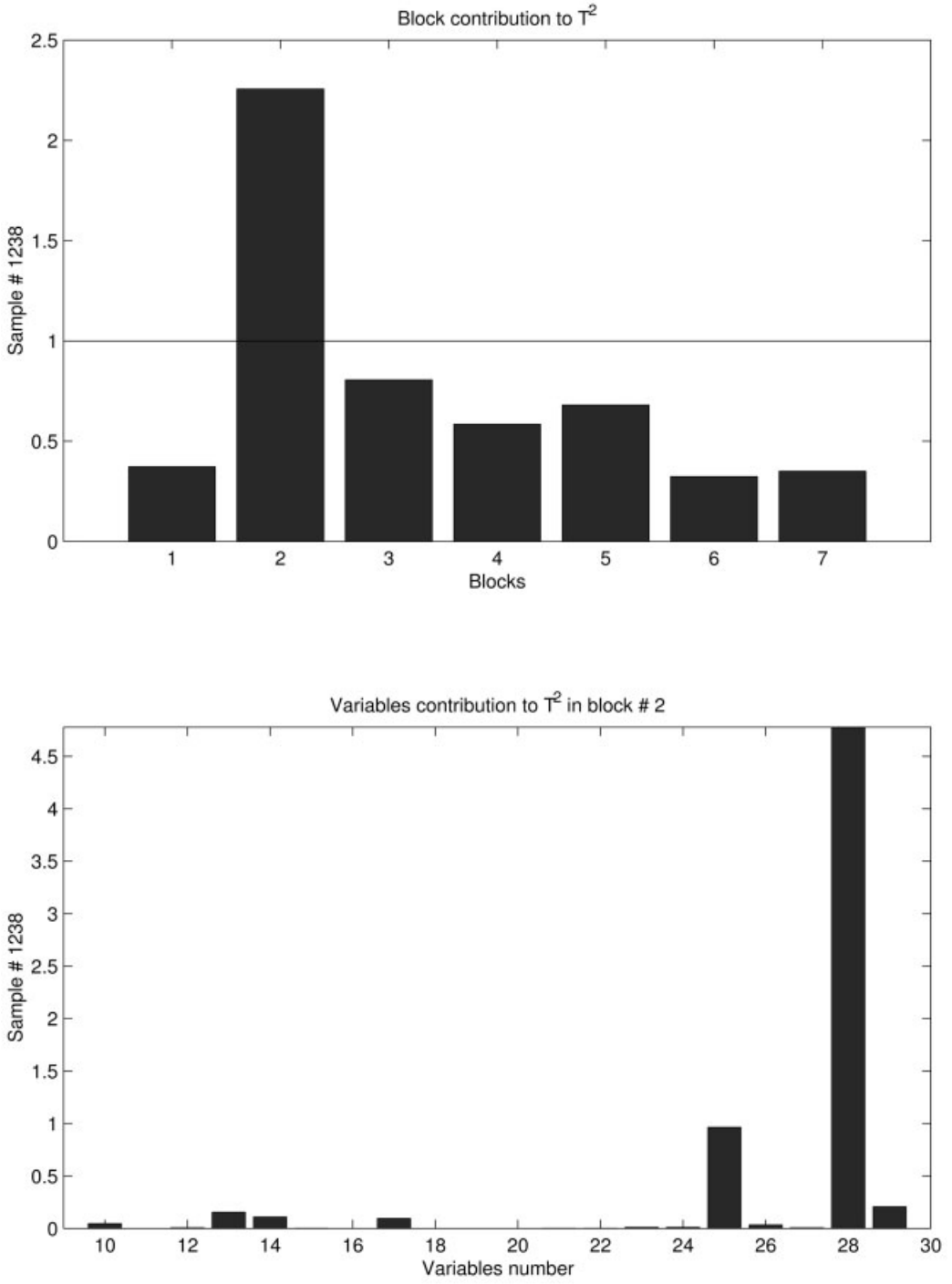


Figure 11. Identification of faulty blocks and contributing variables to T^2.

able or identifiable given the available measurement information and redundancy. Such an analysis can help to determine whether it is possible to carry out the fault detection and diagnosis task or whether additional measurement information needs to be collected. This is analogous to the design of a Kalman filter, where the observability needs to be checked first. Dunia and Qin [12, 13] and Yue and Qin [51] give the necessary and sufficient detectability and identifiability conditions for unidimensional and multidimensional faults. In this section we only give an account of the unidimensional fault analysis, as it is easier to visualize geometrically. The multidimensional fault case can be found in References [13, 51] for the SPE- and combined index-based fault detection and diagnosis.

7.1. Fault detectability

The example process in Figure 1(a) can be used to illustrate the concept of fault detectability: a leakage at the upper stream of sensor x_1 will affect both x_1 and x_2 in exactly the same way and is consistent with the PCA model. This leakage cannot be detected using the SPE index. With the help of a geometric interpretation we give explicit conditions for fault detectability for the unidimensional fault case, where Ξ_i becomes a column vector $\boldsymbol{\xi}_i$. In the presence of a

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process fault $\mathcal{F}_i$ the sample vector $\mathbf{x}$ can be represented using a fault direction vector $\boldsymbol{\xi}_i$:

$$\mathbf{x} = \mathbf{x}^* + f\boldsymbol{\xi}_i \tag{64}$$

where $\boldsymbol{\xi}_i$ is normalized and the scalar f represents the magnitude of the fault $\mathcal{F}_i$. Such a fault belongs to the set of possible faults, denoted by $\{\mathcal{F}_j\}$. This fault vector can represent a sensor fault as well as a process fault.

Since $\boldsymbol{\xi}_i \in \mathbb{R}^m$, it can be projected onto $\mathcal{S}_\mathrm{p}$ and $\mathcal{S}_\mathrm{r}$:

$$\boldsymbol{\xi}_i = \hat{\boldsymbol{\xi}}_i + \tilde{\boldsymbol{\xi}}_i \tag{65}$$

where $\hat{\boldsymbol{\xi}}_i = \mathbf{C}\boldsymbol{\xi}_i \in \mathcal{S}_\mathrm{p}$ and $\tilde{\boldsymbol{\xi}}_i = \tilde{\mathbf{C}}\boldsymbol{\xi}_i \in \mathcal{S}_\mathrm{r}$. If $\tilde{\boldsymbol{\xi}}_i = \mathbf{0}$, the following relation results from Equation (64):

$$\tilde{\mathbf{x}} = \tilde{\mathbf{x}}^* + f\tilde{\boldsymbol{\xi}}_i = \tilde{\mathbf{x}}^*$$

Therefore

$$\mathrm{SPE}(\mathbf{x}) = \mathrm{SPE}(\mathbf{x}^*) \tag{66}$$

As a consequence, no matter how large f is, the fault is not detectable if $\tilde{\boldsymbol{\xi}}_i \neq \mathbf{0}$.

Given that $\tilde{\boldsymbol{\xi}}_i \neq \mathbf{0}$, we define

$$\tilde{\boldsymbol{\xi}}_i^0 \equiv \frac{\tilde{\boldsymbol{\xi}}_i}{||\tilde{\boldsymbol{\xi}}_i||}$$

as the normalized residual direction for the fault vector $\boldsymbol{\xi}_i$. With this notation,

$$\begin{aligned}\tilde{\mathbf{x}} &= \tilde{\mathbf{x}}^* + f\tilde{\boldsymbol{\xi}}_i \\ &= \tilde{\mathbf{x}}^* + \tilde{f}\tilde{\boldsymbol{\xi}}_i^0\end{aligned} \tag{67}$$

where $\tilde{f} = f||\tilde{\boldsymbol{\xi}}_i||$ is the orthogonal distance of the fault to the PCS.

To guarantee that the fault will be detectable, it is required that $\mathrm{SPE} = ||\tilde{\mathbf{x}}||^2 > \delta_\alpha^2$ for all possible normal values of $\mathbf{x}^*$. Figure 12 illustrates geometrically the sufficient condition for detectability in the case of a two-dimensional residual subspace. The vector $\tilde{\mathbf{x}}^*$ can be anywhere inside the circle defined by the upper control limit δ_α. To guarantee that the fault is detectable, the corrupted sample $\tilde{\mathbf{x}}$ has to be outside the circle, which requires that $|\tilde{f}|$ be larger than the diameter of the circle for the extreme case. In other words, one must have

$$|\tilde{f}| > 2\delta_\alpha \tag{68}$$

A general derivation for the case of multidimensional faults can be found in Reference [13].

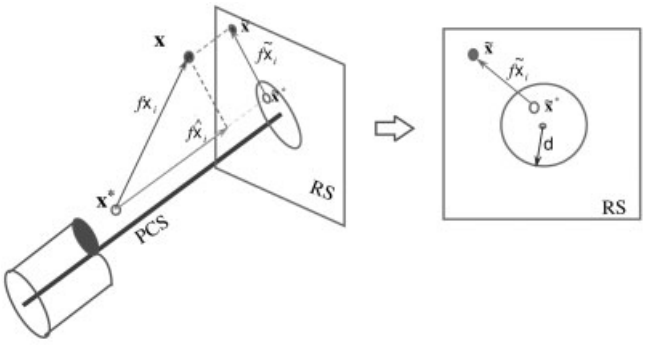


Figure 12. The projection of a faulty sample $\tilde{\mathbf{x}}$ has to be outside the circle defined by the upper control limit for any $\tilde{\mathbf{x}}^*$ inside the circle to guarantee the detection of the fault.

7.2. Fault reconstructability

The feasibility of calculating f_i assures the existence of $\mathbf{x}_i$, which is the best estimate for $\mathbf{x}^*$ by reconstruction in the direction $\boldsymbol{\xi}_i$. Therefore the condition for a feasible calculation of f_i is identical to the condition for reconstructability along $\boldsymbol{\xi}_i$. From Equation (46) it is noticed that f_i can be calculated when $\tilde{\Xi}_i$ (or $\tilde{\boldsymbol{\xi}}_i$) $\neq \mathbf{0}$, i.e. $\boldsymbol{\xi}_i \notin \mathcal{S}_p$. Intuitively, this condition suggests that the displacement caused by $\mathcal{F}_i$ should not lie in the PCS. Therefore a necessary condition for reconstructability is $\tilde{\boldsymbol{\xi}}_i \neq \mathbf{0}$, which is also the necessary condition for detectability.

Dunia and Qin [13] use the variance of reconstruction error (VRE) to measure the reliability of the reconstruction. The VRE in the direction ξ_i is denoted by u_i and represents the variance of the projection of $\mathbf{x}^* - \mathbf{x}_i$ on the fault direction ξ_i:

$$\begin{aligned}u_i &\equiv \mathrm{var}\{\boldsymbol{\xi}_i^\mathrm{T}(\mathbf{x}^* - \mathbf{x}_i)\} \\ &= \frac{\boldsymbol{\xi}_i^\mathrm{T}\mathcal{E}\{\mathbf{x}^*\mathbf{x}^{*\mathrm{T}}\}\tilde{\boldsymbol{\xi}}_i}{(\tilde{\boldsymbol{\xi}}_i^\mathrm{T}\tilde{\boldsymbol{\xi}}_i)^2} = \frac{\tilde{\boldsymbol{\xi}}_i^\mathrm{T}\mathbf{S}\tilde{\boldsymbol{\xi}}_i}{(\tilde{\boldsymbol{\xi}}_i^\mathrm{T}\tilde{\boldsymbol{\xi}}_i)^2}\end{aligned} \tag{69}$$

where $\mathbf{S}$ denotes the covariance matrix of the normal data.

It is possible that the best reconstruction is worse than using the average of the historical data as a reconstruction if a particular sensor is little correlated with other sensors. In other words, it is possible to have

$$u_i > \mathrm{var}\{\boldsymbol{\xi}_i^\mathrm{T}(\mathbf{x} - \bar{\mathbf{x}})\} = \boldsymbol{\xi}_i^\mathrm{T}\mathbf{S}\boldsymbol{\xi}_i \tag{70}$$

In this case the particular sensor or fault is better reconstructed using the historical average instead of the correlation-based PCA model. Dunia and Qin [13] proposed an iterative procedure to determine the number of sensors in the model and the set of faults that can be reliably reconstructed using the PCA model.

Example 4

To illustrate the notion of reconstructability, the process in Figure 2 of five possible faults is considered: $\mathcal{F}_1$ to $\mathcal{F}_3$ are sensor faults, while $\mathcal{F}_4$ and $\mathcal{F}_5$ represent leaks in units A and B. The PCA model for the process with three sensors is

$$\mathbf{p}^\mathrm{T} = \frac{\sqrt{3}}{3}[1 \quad 1 \quad 1]$$

The fault direction for $\mathcal{F}_4$ is

$$\boldsymbol{\xi}_4^\mathrm{T} = \frac{\sqrt{3}}{3}[1 \quad 1 \quad 1]$$

which makes $\tilde{\boldsymbol{\xi}}_4 = \mathbf{0}$. Therefore this fault is neither reconstructable nor detectable. The physical significance is that the fault direction is consistent with the normal variation in the data.

7.3. Fault identifiability

Since the reconstructed SPE is ultimately used for fault identification, it is desirable to have the greatest reduction in SPE when the actual fault $\mathcal{F}_i$ is assumed. However, in some situations the minimized SPE_j for all faults are identical, making them not identifiable.

In general, if the faults $\mathcal{F}_j$ and $\mathcal{F}_i$ are such that $\tilde{\boldsymbol{\xi}}_j^0 = \pm\tilde{\boldsymbol{\xi}}_i^0$, Equations (56) and (57) become

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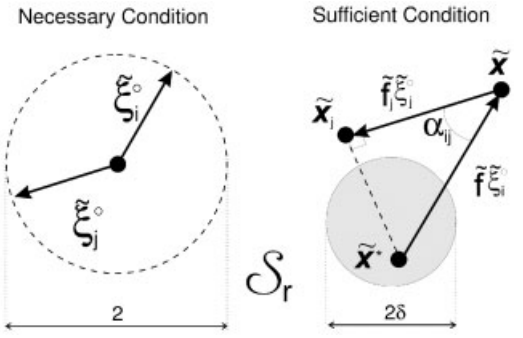


Figure 13. A geometric interpretation of the necessary and sufficient conditions for fault isolability.

$$\begin{aligned}\mathrm{SPE}(\mathbf{x}_j) &= \| \left[\mathbf{I} - \tilde{\xi}_j(\tilde{\xi}_j^{\mathrm{T}}\tilde{\xi}_j)^{-1}\tilde{\xi}_j^{\mathrm{T}}\right]\tilde{\mathbf{x}} \|^2 \\ &= \| \left(\mathbf{I} - \tilde{\xi}_j^0\tilde{\xi}_j^{0\mathrm{T}}\right)\tilde{\mathbf{x}} \|^2 = \| \left(\mathbf{I} - \tilde{\xi}_i^0\tilde{\xi}_i^{0\mathrm{T}}\right)\tilde{\mathbf{x}} \|^2 \\ &= \mathrm{SPE}(\mathbf{x}_i)\end{aligned}$$

Therefore *faults $\mathcal{F}_j$ and $\mathcal{F}_i$ are not identifiable from one another if* $\tilde{\xi}_j^0 = \pm\tilde{\xi}_i^0$. In other words, the fault directions projected on the residual subspace cannot be collinear to be identifiable.

A further question is how large the fault magnitude should be to guarantee identifiability. Dunia and Qin [12] provide the following sufficient condition for fault identifiability:

$$|\tilde{f}| > \frac{2\delta_\alpha}{|\sin\alpha_{ij}|} \tag{71}$$

where α_{ij} represents the angle between $\tilde{\xi}_i$ and $\tilde{\xi}_j$.

A geometric interpretation of the necessary and sufficient conditions for isolability is depicted in Figure 13. The left-hand drawing shows that the fault directions projected on the RS cannot be collinear to be identifiable. The right-hand drawing shows that the fault magnitude $\tilde{f}$ has to be so large that the reconstruction $\tilde{\mathbf{x}}_j$ is guaranteed to be outside the circle, regardless of where $\tilde{\mathbf{x}}^*$ is inside the circle.

Example 5

To illustrate the use of the necessary condition for fault isolation, consider again the process shown in Figure 2. The faults $\mathcal{F}_1$ and $\mathcal{F}_5$ provide the directions

$$\xi_1^{\mathrm{T}} = [1 \quad 0 \quad 0]$$
$$\xi_5^{\mathrm{T}} = \frac{1}{\sqrt{2}}[0 \quad 1 \quad 1]$$

and

$$\tilde{\xi}_1^{0\mathrm{T}} = \frac{\tilde{\xi}_1^{\mathrm{T}}}{||\tilde{\xi}_1||} = \frac{1}{\sqrt{6}}[2 \quad -1 \quad -1]$$

$$\tilde{\xi}_5^{0\mathrm{T}} = \frac{\tilde{\xi}_5^{\mathrm{T}}}{||\tilde{\xi}_5||} = \frac{1}{\sqrt{6}}[-2 \quad 1]$$

are the projections in $\mathcal{S}_r$. These two faults are not isolable, since $\tilde{\xi}_1^0 = -\tilde{\xi}_5^0$. Intuitively, a leak in unit B affects the measurements of the last two sensors, making them inconsistent with the first sensor. This situation is indistinguishable from having a fault in the first sensor.

8. CONCLUSIONS AND DISCUSSION

In this paper we have discussed many basic and advanced issues in statistical process monitoring, including fault detection, identification and reconstruction. Several fault detection indices are unified, including global and subspace-based indices. Fault identification via reconstruction and contribution plots is discussed by analysis and examples. The contribution-based approach has its simplicity and can be generated without prior knowledge. To interpret the contribution plots, process knowledge is needed and should always be used. The reconstruction-based approach requires the knowledge of fault directions, which is readily available for sensor faults and can be extracted from faulty data for process faults [50, 51].

The reconstruction-based approach can also be used without prior knowledge by simply extracting the fault directions and interpreting the fault directions similarly to contribution plots. If a fault has happened once and a fault direction matrix is extracted, the reconstruction-based approach can use the fault direction matrices to identify newly detected faults. This approach is efficient and conclusive for a limited number of process faults that are important to the quality and health of the process. The reconstruction-based approach, with the knowledge of fault directions, provides a framework for fault analysis. Fault detectability, reconstructability and identifiability are discussed in a geometric framework.

If the process to be monitored has a large number of variables, like the polyester film process analyzed in this paper, it can be difficult to interpret the contribution plots. In this case, multiblock analysis [9, 76] and hierarchical contribution plots [70] can be very effective.

While this paper covers many important issues in process monitoring, more advanced methods are available in the literature, including:

- multiway applications [26, 77];
- recursive approaches for adaptive monitoring [22];
- dynamic process monitoring [78–80];
- multiscale approaches using wavelets [19, 25];
- process improvement based on measurements [81].

The analysis and methods presented in this paper can be extended to the topics listed above; see e.g. References [25, 26, 51, 74]. While steady state-based methods work in many practical problems, false alarms can occur if the model is based on steady state data but the process goes through a normal dynamic transient. Another issue is that the normal process is usually time-varying owing to drifts and equipment aging. It is sometimes difficult to distinguish between slow drifts that are normal and incipient process degradation that can lead to more severe abnormal situations. Future effort should be directed to tackling the time-varying and dynamic process behaviors in order to reduce false alarms as well as missing alarms.

While statistical process monitoring has its close connection to multivariate quality control, the shift in the focus on monitoring process variables provides opportunities for deep-level diagnosis as well as challenges for non-stationary process changes and closed-loop feedback effects. Tight

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closed-loop feedback, for example, which is often desired in industrial process operation, can lead to mis-identification of the contributing variables through the feedback loops [53, 82, 83]. McNabb [83] provides a promising approach using PCA models in the feedback-invariant subspace to remove the effect of feedback on fault diagnosis. The presence of feedback and dynamic operations makes it necessary to combine the SPM techniques with dynamic systems theory methods such as Kalman filters and system identification techniques.

Another area that is concerned with the health of process control systems is control performance monitoring and diagnosis [84–87]. The control performance monitoring techniques provide further diagnosis information for process upsets caused by poorly behaved feedback controllers. The integration of control performance monitoring and SPM could lead to further fault discrimination among process faults, sensor faults and control performance-induced upsets. An integrated framework could provide more powerful health monitoring tools and lead to more rapid adoption of these techniques in practice.

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APPENDIX: UPPER CONTROL LIMIT FOR THE COMBINED INDEX φ

The covariance matrix $\mathbf{S}$ is decomposed to

$$\mathbf{S} = \begin{bmatrix}\mathbf{P} & \tilde{\mathbf{P}}\end{bmatrix}\begin{bmatrix}\mathbf{\Lambda} & \mathbf{0} \\ \mathbf{0} & \tilde{\mathbf{\Lambda}}\end{bmatrix}\begin{bmatrix}\mathbf{P} & \tilde{\mathbf{P}}\end{bmatrix}^{\mathrm{T}} = \bar{\mathbf{P}}\begin{bmatrix}\mathbf{\Lambda} & \mathbf{0} \\ \mathbf{0} & \tilde{\mathbf{\Lambda}}\end{bmatrix}\bar{\mathbf{P}}^{\mathrm{T}}$$

where $\bar{\mathbf{P}} = \begin{bmatrix}\mathbf{P} & \tilde{\mathbf{P}}\end{bmatrix}$ is orthogonal. From the definition of φ we have

$$\mathbf{\Phi} = \bar{\mathbf{P}}\mathbf{\Lambda}_{\varphi}\bar{\mathbf{P}}^{\mathrm{T}} = \bar{\mathbf{P}}\begin{bmatrix}\mathbf{\Lambda}^{-1}/\chi^2_{l;\alpha} & \mathbf{0} \\ \mathbf{0} & \mathbf{I}_{m-l/\delta^2_{\alpha}}\end{bmatrix}\bar{\mathbf{P}}^{\mathrm{T}}$$

Therefore

$$\begin{aligned}\mathbf{S\Phi} &= \bar{\mathbf{P}}\begin{bmatrix}\mathbf{\Lambda} & \mathbf{0} \\ \mathbf{0} & \tilde{\mathbf{\Lambda}}\end{bmatrix}\mathbf{\Lambda}_{\varphi}\bar{\mathbf{P}} \\ &= \bar{\mathbf{P}}\begin{bmatrix}\mathbf{I}/\chi^2_{l;\alpha} & \mathbf{0} \\ \mathbf{0} & \tilde{\mathbf{\Lambda}}/\delta^2_{\alpha}\end{bmatrix}\bar{\mathbf{P}}^{\mathrm{T}} \\ (\mathbf{S\Phi})^2 &= \bar{\mathbf{P}}\begin{bmatrix}\mathbf{I}/\chi^4_{l;\alpha} & \mathbf{0} \\ \mathbf{0} & \tilde{\mathbf{\Lambda}}^2/\delta^4_{\alpha}\end{bmatrix}\bar{\mathbf{P}}^{\mathrm{T}}\end{aligned}$$

and

$$\begin{aligned}\mathrm{tr}(\mathbf{S\Phi}) &= \frac{l}{\chi^2_{l;\alpha}} + \frac{\sum_{i=l+1}^{m}\lambda_i}{\delta^2_{\alpha}} \\ \mathrm{tr}(\mathbf{S\Phi})^2 &= \frac{l}{\chi^4_{l;\alpha}} + \frac{\sum_{i=l+1}^{m}\lambda_i^2}{\delta^4_{\alpha}}\end{aligned}$$

REFERENCES

1. Wise BM, Veltkamp DJ, Davis B, Ricker NL, Kowalski BR. Principal component analysis for monitoring the West Valley liquid fed ceramic melter. *Waste Management'88 Proc.* 1988; 811–818.
2. MacGregor JF. Multivariate statistical methods for monitoring large datasets from chemical processes. *AIChE Meet.*, San Francisco, CA, 1989.
3. Wise BM, Ricker NL. Feedback strategies in multiple sensor systems. *AIChE Symp. Ser. Process Sens.* 1989; **85**: 19–23.
4. Kresta JV, MacGregor JF, Marlin TE. Multivariate statistical monitoring of processes. *Can. J. Chem. Eng.* 1991; **69**(1): 35–47.
5. Jackson JE. *A User's Guide to Principal Components*. Wiley-Interscience: New York, 1991.
6. Wise BM, Ricker NL. Recent advances in multivariate statistical process control: improving robustness and sensitivity. *Proc. IFAC ADCHEM Symp.*, 1991; 125–130.
7. Piovoso MJ, Kosanovich KA, Yuk JP. Process data chemometrics. *IEEE Trans. Instrum. Meas.* 1992; **41**: 262–268.
8. Miller P, Swanson RE, Heckler CF. Contribution plots: the missing link in multivariate quality control. *Fall Conf. of ASQC and ASA*, Milwaukee, WI, 1993.
9. MacGregor JF, Jaeckle C, Kiparissides C, Koutoudi M. Process monitoring and diagnosis by multiblock PLS methods. *AIChE J.* 1994; **40**: 826–828.
10. Dunia R, Qin SJ, Edgar TF, McAvoy TJ. Identification of faulty sensors using principal component analysis. *AIChE J.* 1996; **42**: 2797–2812.
11. Raich AC, Cinar A. Process disturbance diagnosis by statistical distance and angle measures. *Proc. IFAC Congr.*, vol. N, San Francisco, CA, 1996; 283–288.
12. Dunia R, Qin SJ. A unified geometric approach to process and sensor fault identification: the unidimensional fault case. *Comput. Chem. Eng.* 1998; **22**: 927–943.
13. Dunia R, Qin SJ. Subspace approach to multidimensional fault identification and reconstruction. *AIChE J.* 1998; **44**: 1813–1831.
14. Yue H, Qin SJ. Fault reconstruction and identification for industrial processes. *AIChE Ann. Meet.*, Miami, FL, 1998.
15. Qin SJ, Li W. Detection, identification and reconstruction of faulty sensors with maximized sensitivity. *AIChE J.* 1999; **45**: 1963–1976.
16. Gertler J, Li W, Huang Y, McAvoy TJ. Isolation-enhanced principal component analysis. *AIChE J.* 1999; **45**(2): 323–334.
17. Stork CL, Kowalski BR. Distinguishing between process upsets and sensor malfunctions using sensor redundancy. *Chemometrics Intell. Lab. Syst.* 1999; **46**: 117–131.
18. Wise BM, Gallagher NB, Butler SW, White D, Barna GG. Development and benchmarking of multivariate statistical process control tools for a semiconductor etch process: impact of measurement selection and data treatment on sensitivity. *IFAC SAFEPROCESS '97*, Hull, 1997.
19. Bakshi BR. Multiscale PCA with application to multivariate statistical process monitoring. *AIChE J.* 1998; **44**: 1596–1610.
20. Dayal BS, MacGregor JF. Recursive exponentially weighted PLS and its applications to adaptive control and prediction. *J. Process Control* 1997; **7**: 169–179.
21. Qin SJ. Recursive PLS algorithms for adaptive data modeling. *Comput. Chem. Eng.* 1998; **23**: 503–514.
22. Li W, Yue H, Valle-Cervantes S, Qin SJ. Recursive PCA for adaptive process monitoring. *J. Process Control* 2000; **10**: 471–486.
23. Ku W, Storer RH, Georgakis C. Disturbance detection and isolation by dynamic principal component analysis. *Chemometrics Intell. Lab. Syst.* 1995; **30**: 179.

Copyright © 2003 John Wiley & Sons, Ltd.

24. Negiz A, Cinar A. Statistical monitoring of multivariate dynamic processes with state space models. *AIChE J.* 1997; **43**: 2002–2020.
25. Misra M, Qin SJ, Yue H, Ling C. Multivariate process monitoring and fault identification using multi-scale PCA. *Comput. Chem. Eng.* 2002; **26**: 1281–1293.
26. Nomikos P, MacGregor JF. Multivariate SPC charts for monitoring batch processes. *Technometrics* 1995; **37**: 41–59.
27. Mah RSH, Stanley GM, Downing D. Reconciliation and rectification of process ow and inventory data. *Ind. Eng. Chem. Process Design Develop* 1976; **15**: 175.
28. Romagnoli JA, Stephanopoulos G. Rectification of process measurement data in the presence of gross errors. *Chem. Eng. Sci.* 1981; **36**: 1849–1863.
29. Narasimhan S, Mah RSH. Generalized likelihood ratios for gross error identification. *AIChE J.* 1987; **33**: 1514–1521.
30. Narasimhan S, Mah RSH. Generalized likelihood ratios for gross error identification in dynamic processes. *AIChE J.* 1988; **34**: 1321–1331.
31. Chow EY, Willsky AS. Analytical redundancy and the design of robust failure detection systems. *IEEE Trans. Automatic Control* 1984; **29**: 603–614.
32. Isermann R. Process fault detection based on modeling and estimation methods—a survey. *Automatica* 1984; **20**: 387–404.
33. Gertler J. Survey of model-based failure detection and isolation in complex plants. *IEEE Control Syst. Mag.* 1988; **12**: 3–11.
34. Benveniste A, Basseville M, Moustakides G. The asymptotic local approach to change detection and model validation. *IEEE Trans. Automatic Control* 1987; **32**: 583–592.
35. Frank PM. Fault diagnosis in dynamic systems using analytical and knowledge-based redundancy—a survey and some new results. *Automatica* 1990; **26**: 459–474.
36. Frank PM. Analytical and qualitative model-based fault diagnosis—a survey and some new results. *Eur. J. Control* 1996; **2**: 6–28.
37. Cleason TC, Staelin R. A proposal for handling missing data. *Psychometrika* 1975; **40**: 229–252.
38. Martens H, Naes T. *Multivariate Calibration*. Wiley: New York, 1989.
39. Nelson PRC, Taylor PA, MacGregor JF. Missing data methods in PCA and PLS: score calculations with incomplete observations. *Chemometrics Intell. Lab. Syst.* 1996; **35**: 45–65.
40. Kourti T, MacGregor JF. Multivariate SPC methods for monitoring and diagnosing of process performance. *Proc. PSE*, 1994; 739–746.
41. Tong H, Crowe CM. Detection of gross errors in data reconciliation by principal component analysis. *AIChE J.* 1995; **41**: 1712–1722.
42. Kramer MA. Autoassociative neural networks. *Comput. Chem. Eng.* 1992; **16**: 313–328.
43. MacGregor JF. Statistical process control of multivariate processes. *Prepr. IFAC ADCHEM*, 1994.
44. MacGregor JF, Kourti T. Statistical process control of multivariate processes. *Control Eng. Pract.* 1995; **3**: 403–414.
45. Wise BM, Gallagher NB. The process chemometrics approach to process monitoring and fault detection. *J. Process Control* 1996; **6**: 329–348.
46. Chiang LH, Russell EL, Braatz RD. *Fault Detection and Diagnosis in Industrial Systems*. Springer: London, 2001.
47. Wise BM, Ricker NL, Veltkamp DF, Kowalski BR. A theoretical basis for the use of principal component models for monitoring multivariate processes. *Process Control Qual.* 1990; **1**: 41–51.
48. Wold S, Esbensen K, Geladi P. Principal component analysis. *Chemometrics Intell. Lab. Syst.* 1987; **2**: 37–52.
49. Yoon S, MacGregor JF. Fault diagnosis with multivariate statistical models. Part I: using steady state fault signatures. *J. Process Control* 2001; **11**: 387–400.
50. Valle-Cervantes S, Qin SJ, Piovoso MJ, Bachmann M, Mandakoro N. Extracting fault subspaces for fault identification of a polyester film process. *Proc. ACC*, Arlington, VA, 2001; 4466–4471.
51. Yue H, Qin SJ. Reconstruction based fault identification using a combined index. *Ind. Eng. Chem. Res.* 2001; **40**: 4403–4414.
52. Ljung L. *System Identification: Theory for the User*. Prentice-Hall: Englewood Cliffs, NJ, 1999.
53. Yoon S, MacGregor JF. Statistical and causal model-based approaches to fault detection and isolation. *AIChE J.* 2000; **46**: 1813–1824.
54. Li W, Qin SJ. Consistent dynamic PCA based on errors-in-variables subspace identification. *J. Process Control* 2001; **11**: 661–678.
55. Raich A, Cinar A. Statistical process monitoring and disturbance diagnosis in multivariate continuous processes. *AIChE J.* 1996; **42**: 995–1009.
56. Jackson JE, Mudholkar G. Control procedures for residuals associated with principal component analysis. *Technometrics* 1979; **21**: 341–349.
57. Box GEP. Some theorems on quadratic forms applied in the study of analysis of variance problems. I. Effect of inequality of variance in the one-way classification. *Ann. Math. Statist.* 1954; **25**: 290–302.
58. Tracy ND, Young JC, Mason RL. Multivariate control charts for individual observations. *J. Qual. Technol.* 1992; **24**: 88–95.
59. Hawkins DM. The detection of errors in multivariate data using principal components. *J. Am. Statist. Assoc.* 1974; **69**: 340–344.
60. Mardia KV. Mahalanobis distances and angles. In *Multivariate Analysis—IV*, Krishnaiah PR (ed.). North-Holland: Amsterdam, 1977; 495–511.
61. Takemura A. A principal decomposition of Hotelling's T^2 statistic. In *Multivariate Analysis—VI*, Krishnaiah PR (ed.). North-Holland: Amsterdam, 1985; 583–597.
62. Wold H. Nonlinear estimation by iterative least squares procedures. In *Research Papers in Statistics*, David F (ed.). Wiley: New York, 1966.
63. Qin SJ, Dunia R. Determining the number of principal components for best reconstruction. *Proc. 5th IFAC DYCOPS*, Corfu, 1998; 359–364.
64. Wold S. Cross validatory estimation of the number of components in factor and principal component analysis. *Technometrics* 1978; **20**: 397–406.
65. Kano M, Nagao K, Hasebe S, Hashimoto I, Ohno H. Statistical process monitoring based on dissimilarity of process data. *AIChE J.* 2002; **48**: 1231–1240.
66. Chiang LH, Russell EL, Braatz RD. Fault diagnosis and Fisher discriminant analysis, discriminant partial least squares, and principal component analysis. *Chemometrics Intell. Lab. Syst.* 2000; **50**: 243–252.
67. Kourti T, MacGregor JF. Multivariate SPC methods for process and product monitoring. *J. Qual. Technol.* 1996; **28**: 409–428.
68. Hopkins RW, Miller P, Swanson RE, Scheible JJ. Method of controlling a manufacturing process using multivariate analysis. *US Patent 5442562*, 1995.
69. Nomikos P. Statistical monitoring of batch processes. *Prepr. Joint Statistical Meet.*, Anaheim, CA, 1997.
70. Qin SJ, Valle-Cervantes S, Piovoso M. On unifying multiblock analysis with applications to decentralized process monitoring. *J. Chemometrics* 2001; **15**: 715–742.
71. Westerhuis JA, Gurden SP, Smilde AK. Generalized contribution plots in multivariate statistical process monitoring. *Chemometrics Intell. Lab. Syst.* 2000; **51**: 95–114.

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72. Conlin AK, Martin EB, Morris AJ. Confidence limits for contribution plots. *J. Chemometrics* 2000; **14**: 725–736.
73. Runger GC, Alt FB, Montgomery DC. Contributors to a multivariate statiscal process control chart signal. *Commun. Statist.—Theory Methods* 1996; **25**: 2203–2213.
74. Cherry G, Good R, Qin SJ. Semiconductor process monitoring and fault detection with recursive multiway PCA based on a combined index. *AEC/APC Symp. XIV*, Salt Lake City, UT, 2002.
75. Valle-Cervantes S. Plant-wide monitoring of processes under closed-loop control. *PhD Thesis*, University of Texas at Austin, 2001.
76. Westerhuis JA, Kourti T, MacGregor JF. Analysis of multiblock and hierarchical PCA and PLS models. *J. Chemometrics* 1998; **12**: 301–321.
77. Smilde A. Comments on three-way analyses used for batch process data. *J. Chemometrics* 2001; **15**: 15–27.
78. Wang Y, Seborg D, Larimore W. Process monitoring based on canonical variate analysis. *Proc. ADCHEM 97*, Banff, 1997; 523–528.
79. Qin SJ, Li W. Detection and identification of faulty sensors in dynamic processes. *AIChE J.* 2001; **47**: 1581–1593.
80. Woodall WH, Montgomery DC. Research issues and ideas in statistical process control. *J. Qual. Technol.* 1999; **31**: 376–386.
81. Bonvin D, Srinivasan B, Ruppen D. Dynamic optimization in the batch chemical industry. *Prepr. Chemical Process Control-6, Assessment and New Directions for Research (CPC-6)*, Tuscon, AZ, 2001.
82. Pranatyasto TN, Qin SJ. Sensor validation and process fault diagnosis for FCC units under MPC feedback. *Control Eng. Pract.* 2001; **9**: 877–888.
83. McNabb CA. MIMO control performance monitoring based on subspace projections. *PhD Thesis*, University of Texas at Austin, 2002.
84. Qin SJ. Control performance monitoring—a review and assessment. *Comput. Chem. Eng.* 1998; **23**: 178–186.
85. Harris TJ, Seppala CT. Recent developments in controller performance monitoring and assessment techniques. *Chemical Process Control—CPC VI*, Tuscon, AZ, 2002; 208–222.
86. Kozub DJ. Controller performance monitoring and diagnosis: experiences and challenges. *Fifth Int. Conf. on Chemical Process Control*, Tahoe, CA, 1996; 83–96.
87. Harris TJ, Seppala CT, Desborough LD. A review of performance monitoring and assessment techniques for univariate and multivariate control systems. *J. Process Control* 1999; **9**: 1–17.

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